Heaven's Light is Our guide

Rajshahi University of Engineering & Technology

Department of Industrial and Production Engineering

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Group no: 11

Title: A Wheelchair-Bed that Converts to a Bed with Cleaning Feature for Improved Mobility and Comfort for Immobile Patients

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Their knowledgeable direction, insightful criticism, and constant encouragement were crucial in helping us reach precise and significant outcomes. We truly value their commitment and the time they generously sacrificed during crucial points in our project.

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In essence, our project's success is a result of our team's combined effort and the mentors' direction, rather than just the sum of our individual efforts. We feel honoured to join together in celebrating this achievement.

Summary

The objective of our study is to create a wheelchair that can be turned into a bed and has a self-cleaning feature. The purpose of this study is to evaluate consumer demand for wheelchairs with these cutting-edge technologies, which are intended to improve immobile patients' comfort and mobility.

We are developing a wheelchair-bed hybrid that easily converts into a bed and has a cleaning function to enhance patient care based on input from customers. We carried out an extensive market survey to inform our design process, and the results showed a high level of satisfaction, indicating that there is a need for these kinds of improvements.

We determined the essential features that users needed by using the Kano model analysis. After that, we used Quality Function Deployment (QFD) to make sure our design would completely satisfy those requirements. To further improve our product, we also thoroughly compared our design to those already on the market.

A Quality Function Development(QFD) analysis, which provides a thorough dissection of the product's structure and functionalities, is also included in the report. The convertible wheelchair with self-cleaning technology will be manufactured using the 2D, 3D, and exploded views that we have provided as part of the finished design phase.

Chapter 01: Introduction

1.1 Insight

The demand for advanced assistive devices such as wheelchairs has become increasingly vital in providing improved mobility and comfort for individuals with disabilities, particularly in Bangladesh. Patients who are unable to move independently require sophisticated solutions that go beyond traditional wheelchair designs.

In response to this, innovation in wheelchair design has progressed, with a focus on integrating functionalities that address both physical limitations and the demand for greater convenience. A key insight from this study is the necessity for features that not only assist with mobility but also enhance comfort and accessibility, areas that current products often fail to meet.

1.2 Previous Iterations

Earlier designs of wheelchairs focused mainly on helping people move, but they didn't always consider user comfort or extra features for daily care. Many of the available models are not adaptable enough for different medical needs, especially for those needing hygiene or resting assistance. Some wheelchairs have improved in areas like ease of movement, but they still lack important features like reclining seats and built-in commode systems. These missing features are especially common in the Bangladeshi market.

1.3 Our Expectations

Our proposed design aims to combine a user-friendly defecation system with an ergonomic reclining system, along with supportive arm and head rests, ensuring maximum comfort and ease of use. By incorporating these features, we expect to create a wheelchair that not only enhances mobility but also addresses critical aspects of comfort, dignity, and independence. We hope this design will set a new standard for multi-functional wheelchairs, improving the overall quality of life for users.

Chapter 02: Market Analysis

2.1 Need Statement

The rising focus on improving the quality of life for individuals with limited mobility has led to a growing demand for innovative and user-friendly solutions in assistive devices. Many existing wheelchair designs provide basic functionality but lack features that address essential needs like hygiene, comfort, and ease of movement. Our multi-functional wheelchair is designed to provide a comprehensive solution by incorporating a defecation system, reclining mechanism, and supportive arm and headrests. This will significantly reduce the discomfort faced by users in managing everyday activities, particularly hygiene-related tasks, and will enhance independence. Additionally, the device will be portable and affordable, making it accessible to a wide range of users.

2.2 Potential Customer

Our target audience for this wheelchair includes individuals who require mobility assistance due to physical disabilities or age-related mobility challenges. This product will particularly benefit:

- **Individuals with limited mobility**, including elderly people, who need additional support for hygiene and comfort.
- **Caregivers** looking for easier solutions to assist people with reduced mobility.
- **Hospitals, nursing homes, and rehabilitation centers** that require advanced wheelchairs for patients in need of extra assistance.

2.3 Target Audience

Our target audience for the multi-functional wheelchair includes individuals of all ages who require mobility assistance, particularly those with physical disabilities, age-related mobility challenges, or chronic health conditions. These individuals often face daily challenges in managing their hygiene and comfort, and they seek a solution that offers both independence and dignity.

2.4 Market Trend

An ageing population and rising demand for cutting-edge assistive technology are driving the wheelchair market's rapid expansion. Innovative designs that improve comfort, mobility, and independence—like bed-wheelchairs that can be converted into wheelchairs and versions with self-cleaning capabilities—are becoming more and more popular. Customers are looking for multipurpose goods that can be used in both home and medical settings, providing convenience and better treatment for patients who are unable to move around. As manufacturers place a higher priority on user-friendly designs, technological innovations like automation and smart features are becoming more popular. The need for advanced wheelchairs is also being driven by the increased emphasis on healthcare infrastructure and individualized care solutions

2.5 Customer Requirement

Customer requirements for our wheelchair focus on addressing the needs of individuals with limited mobility. These include features like an integrated defecation system for hygiene, a reclining mechanism for comfort, and portability.

Serial No	Requirements	Importance (1-10)
1	180 Degree Decline	9
2	Defecation System	9
3	Comfort	8
4	Leg, Arm and Head Rest	7
5	Cup holder	4
6	Cleaning System	8
7	Low Cost	8
8	Lightweight	8
9	Safety Feature	7
10	Accessibility features	7

Table 4.1 Customer Requirement

2.6 Objective Tree

The objective tree for wheelchair design provides a structured framework to ensure the creation of an efficient, comfortable, and accessible wheelchair. By breaking down the main objective into primary goals—comfort, accessibility, durability, mobility, safety, and affordability—we ensure all critical aspects are addressed. This comprehensive approach guides designers and engineers in developing wheelchairs that enhance user independence and quality of life. The objective tree highlights the importance of considering diverse user needs, ultimately contributing to a more inclusive and supportive mobility solution.

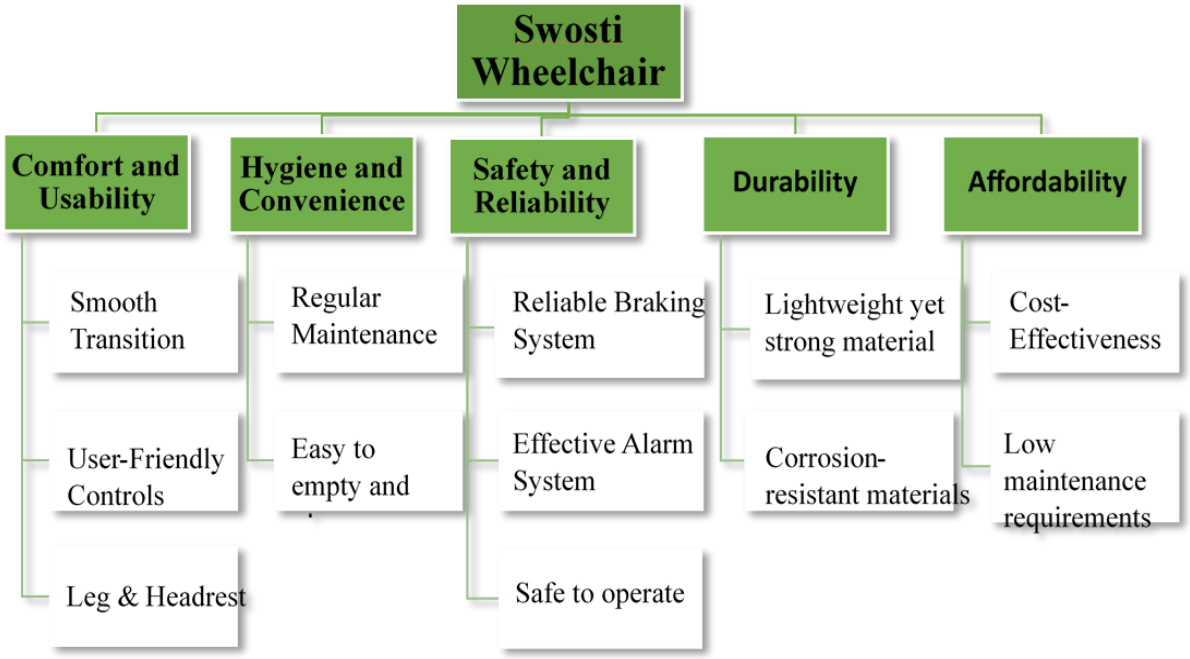


Figure 4.1 Objective Tree

Chapter 03: Customer Survey

3.1 Description

A customer survey is an investigative instrument that gathers input from consumers to assist businesses in comprehending their contentment, anticipations, and involvement. The designer uses this survey to gather information about the product and design in accordance with client specifications. Understanding the demands of the consumer is essential because it allows us to determine whether or not the product will be lucrative. We conducted a customer survey on "A Wheelchair-Bed that Converts to a Bed with Self-Cleaning Feature for Improved Mobility and Comfort for Immobile Patients". We have received in total of 105 responses from diverse communities. In this report, we will discuss the outcome of our customer survey.

3.2 Objectives

1. Evaluate user preferences and needs
2. Evaluate potential acceptance
3. Identify key features and priorities
4. Get feedback on usability and design
5. Assess overall satisfaction level

3.3 Data Analysis

Age & Occupation: We have recorded age & occupation of the customer to get the demographics of the customer. It helps us to understand Market segmentation. We get to aware of our target customer. Mainly our target customer are aged between 45 to 60 and 60+.

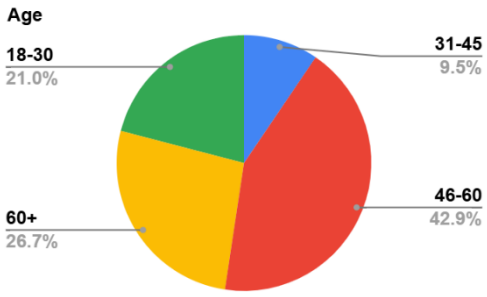


Figure 5.1 Respondents According to age

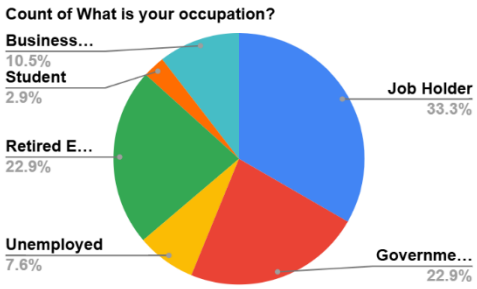


Figure 5.2 Respondents According to Occupation

Use of a wheelchair by our reviewer: 83.8% of our customers are either wheelchair users directly or have had a relative use one. Thus, the information we have gathered is relevant. Their evaluation will be sincere

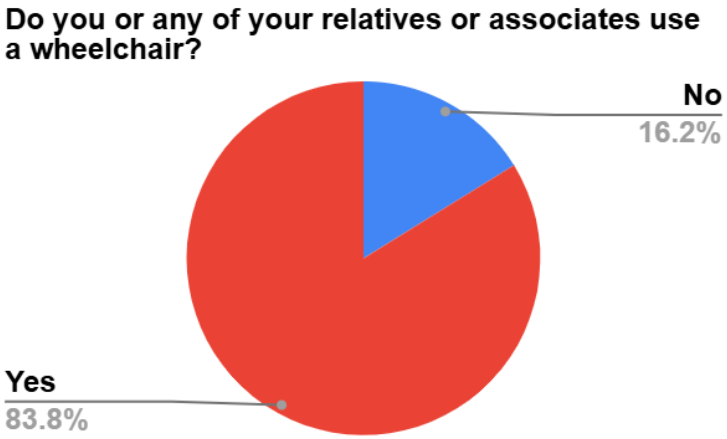


Figure 5.3 Respondents According to users of Wheelchair

Problem Regarding Current Wheelchairs: Here we tried to find out the what are the current problems customers are facing with existing chairs. From the survey we have found that lack of accessibility features and discomfort are the major problems of current wheelchairs. From the survey 33% of the respondents face Discomfort, 27.3% face Difficulty Maneuvering and 39.8% face Lack of Accessibility features problems

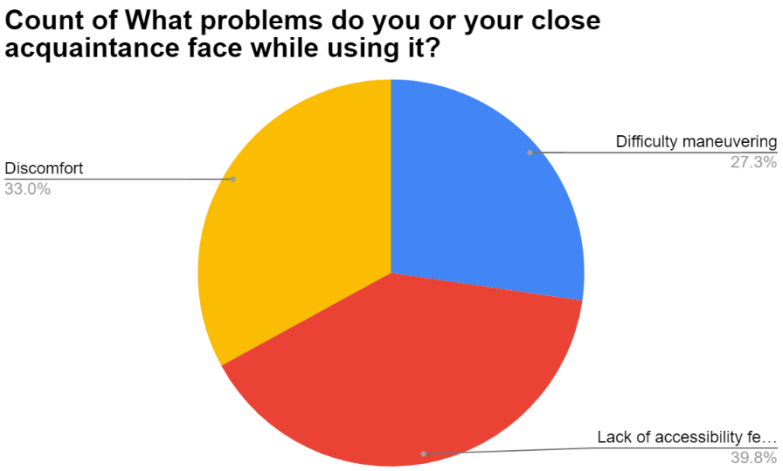


Figure 5.4 Respondents According to the problems faced by existing Wheelchair

Insight: The key insight from this pie chart is most of people feel the necessity of lack of accessibility features & discomfort.

Satisfaction with the current wheelchair: We have found that current wheelchairs are not able to satisfy the current customer. They demand few modifications on current designs. Here numbering 1 is unsatisfied and 5 is highly satisfied. Here we can see maximum consumers are not fully satisfy with the design, adjustability and accessibility features of the existing wheelchairs.

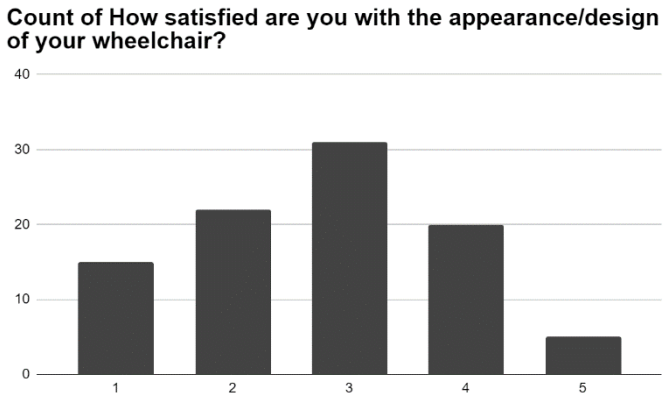


Figure 5.5 Respondents Satisfaction rate with appearance/ design

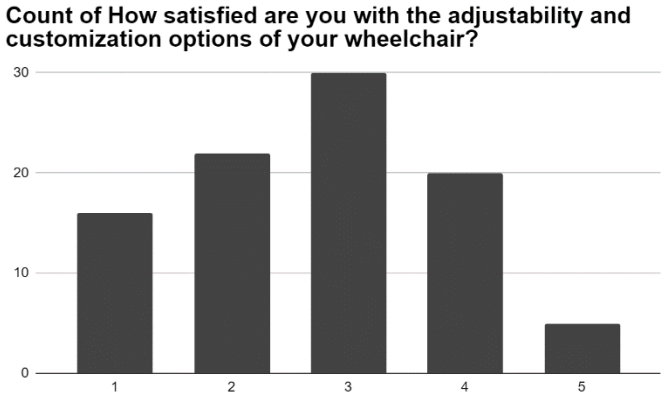


Figure 5.6 Respondents Satisfaction rate with adjustability and customization

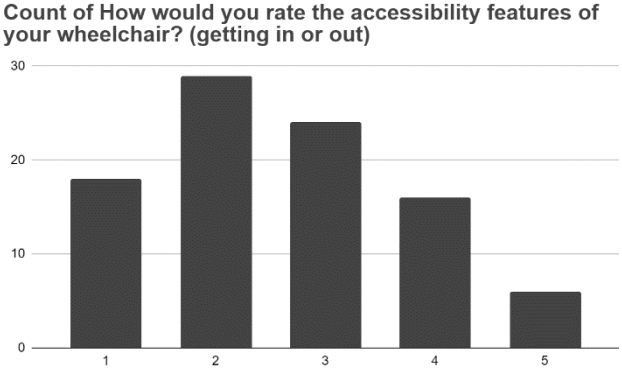


Figure 5.7 Respondents Satisfaction rate with the feature getting in and out

Insight: The above pie chart(Fig 3.4, Fig 3.5, Fig 3.6) demonstrates that most people are not satisfied with the current wheelchairs which are available in the market.

Features and Rating of Our Proposed Product: From the rating by our target audience we can know what actually the customers want from the product and find customers mostly want cleaning systems & 180 degree conversations of the wheelchairs. Here numbering 1 is unsatisfied and 5 is highly satisfied.

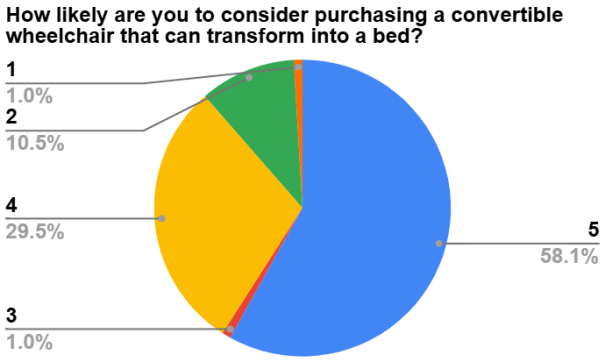


Figure 5.8 Respondents Satisfaction rate with the feature that transforms into bed

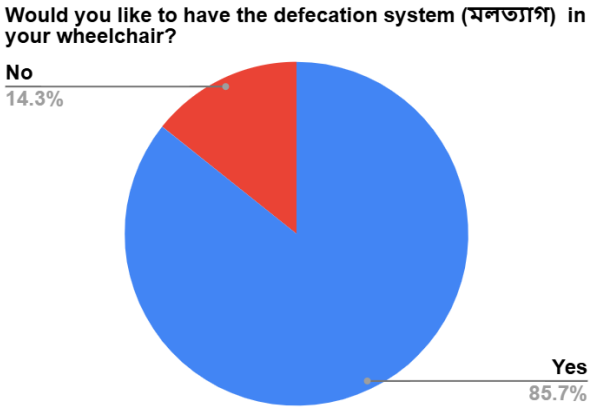


Figure 5.9 Respondents Satisfaction rate with the feature defecation system

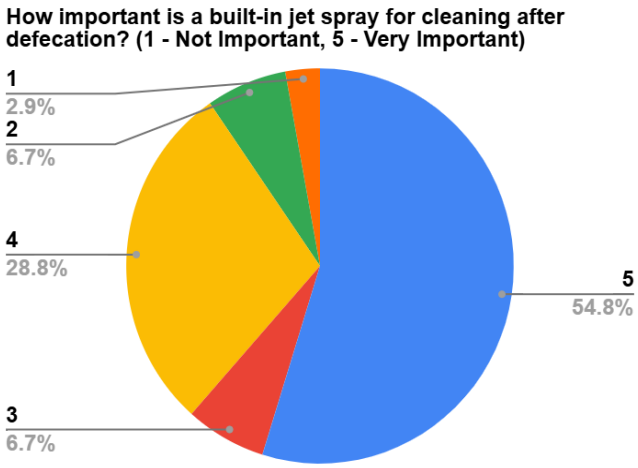


Figure 5.10 Respondents Satisfaction rate with the feature build-in jet spray

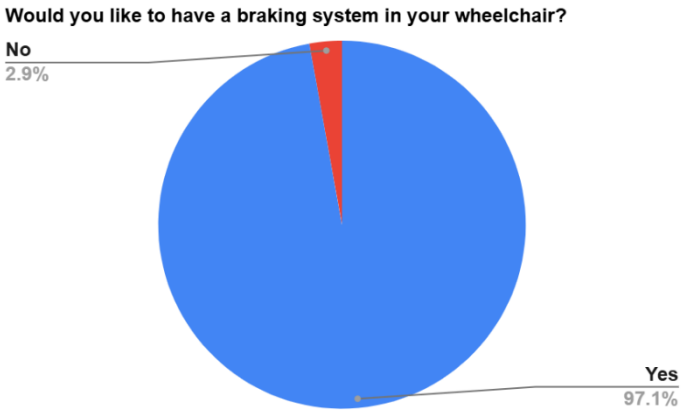


Figure 5.11 Respondents Satisfaction rate with the feature build-in jet spray

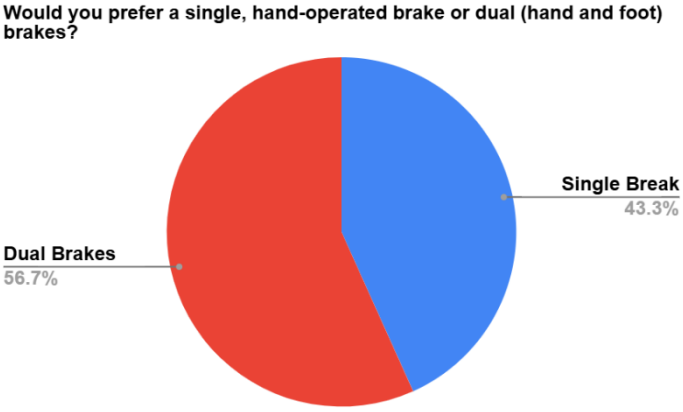


Figure 5.12 Respondents about dual/single brake

Insight: Figures 3.7 to 3.11 illustrate our understanding of customer needs and provide valuable insights into consumer perspectives, which were crucial in shaping the design and development of our wheelchair. These findings served as key components in creating our desired product.

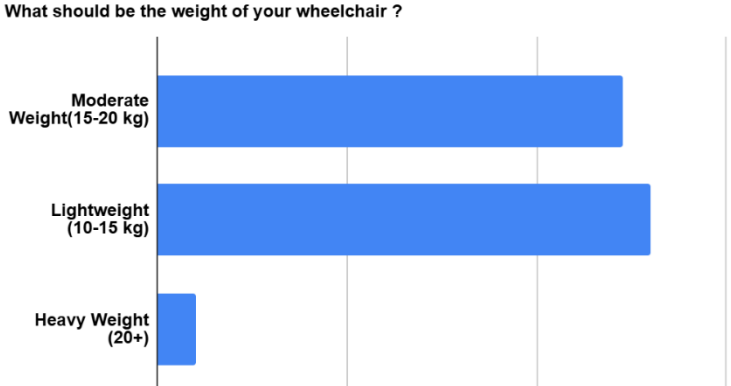


Figure 5.13 Respondents about weight of Wheelchair

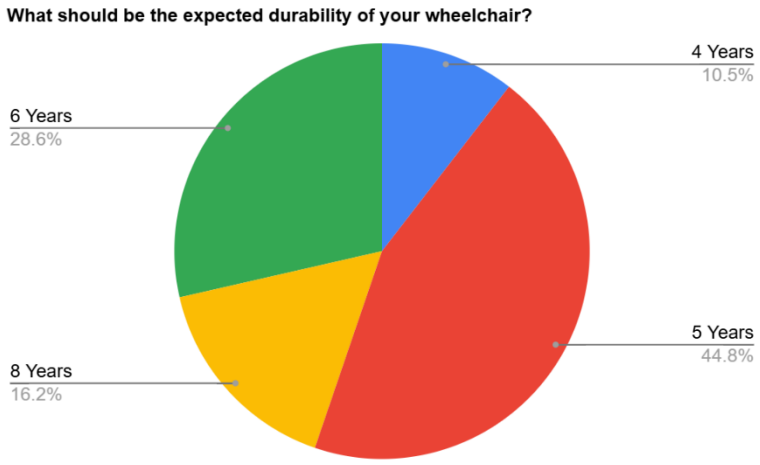


Figure 5.14 Respondents about durability of wheelchair

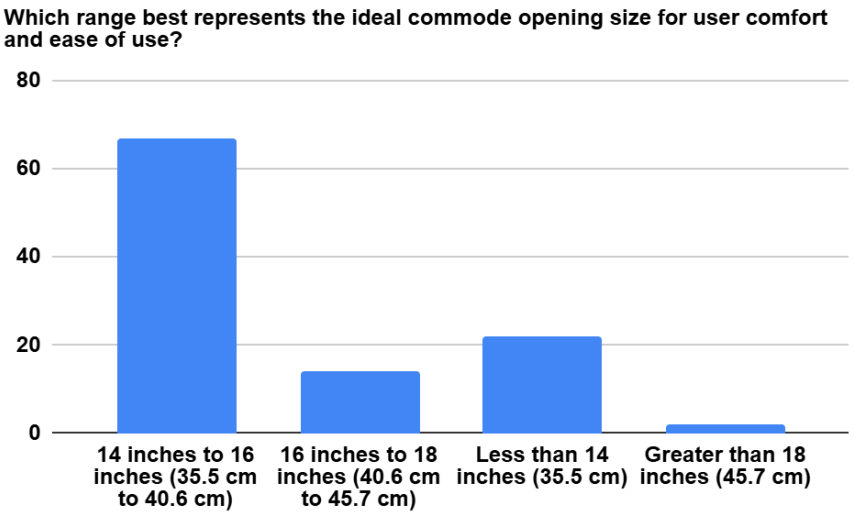


Figure 5.15 Respondents about the dimension of the commode

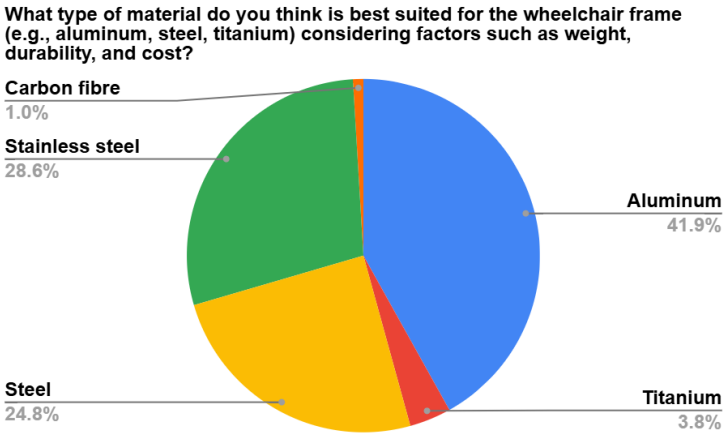


Figure 5.16 Respondents about the material of wheelchair

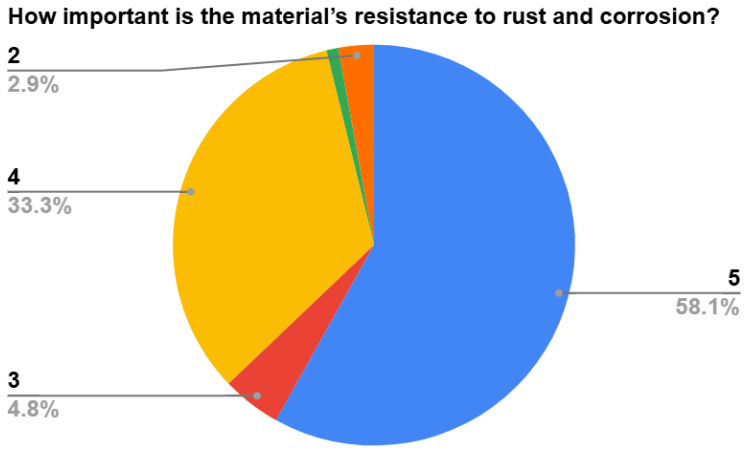


Figure 5.17 Respondents about the material's resistance to rust and corrosion

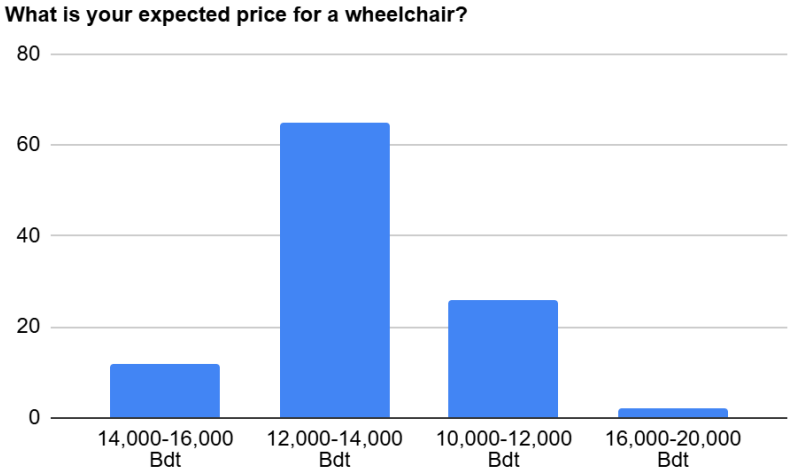


Figure 5.18 Respondents about the price of a wheelchair

Insight: Figures 3.12 to 3.17 include essential details about the elements of cost, materials, weight, and durability. To guarantee accessibility and affordability for a wide range of users, the wheelchair needs to be lightweight, affordable, and durable, according to the data from these graphs.

Rating of our Features: We are trying to develop five features at our wheelchair. These are 180 degree decline, defecation system, leg, arm and headrest, cup holder and cleaning system. Now we will see the feedback of our integrated features. Here numbering 1 is unsatisfied and 5 is highly satisfied.

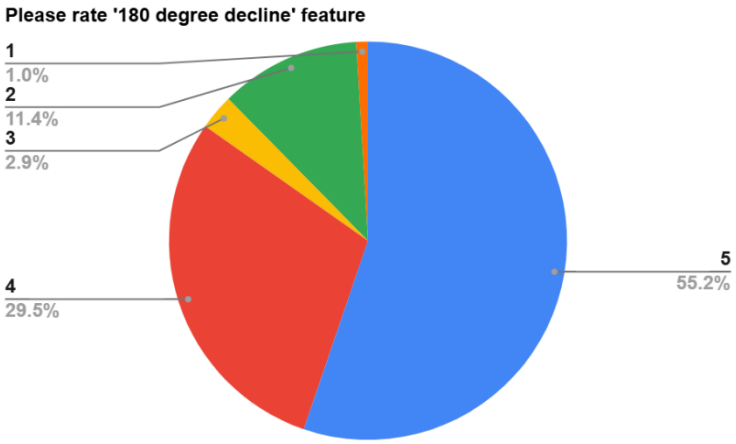


Figure 5.19 Respondents about the features of 180 degree decline

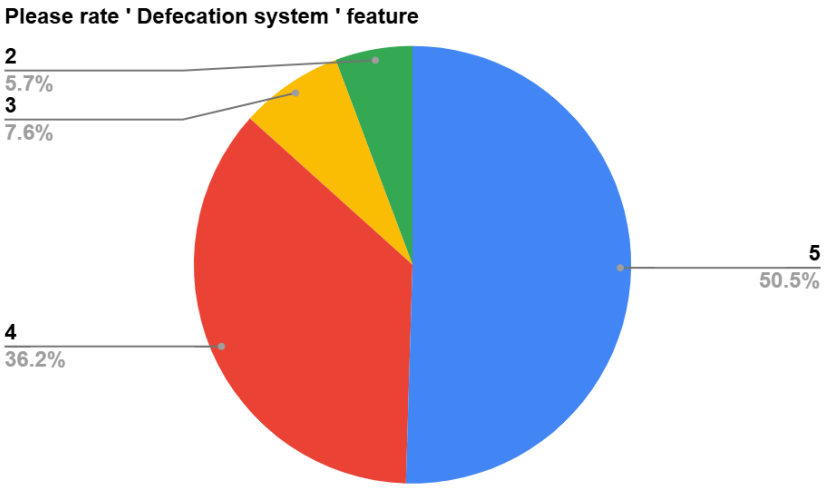


Figure 5.20 Respondents about the features of defecation system

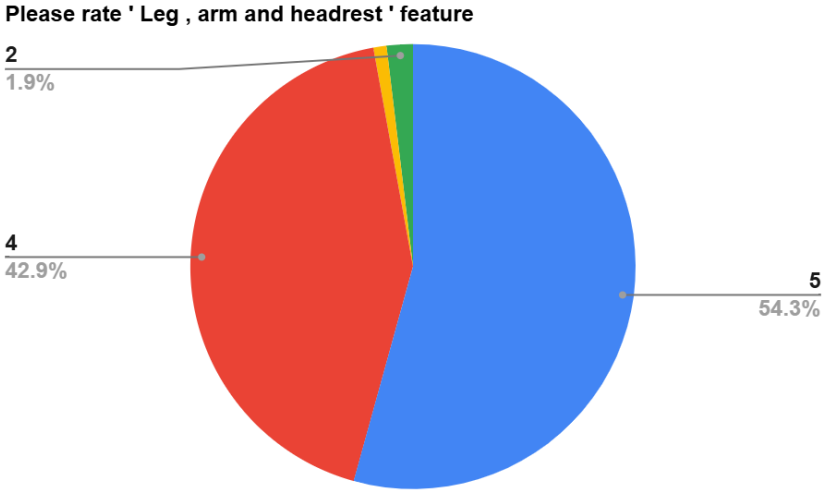


Figure 5.21 Respondents about the features of leg, arm and headrest

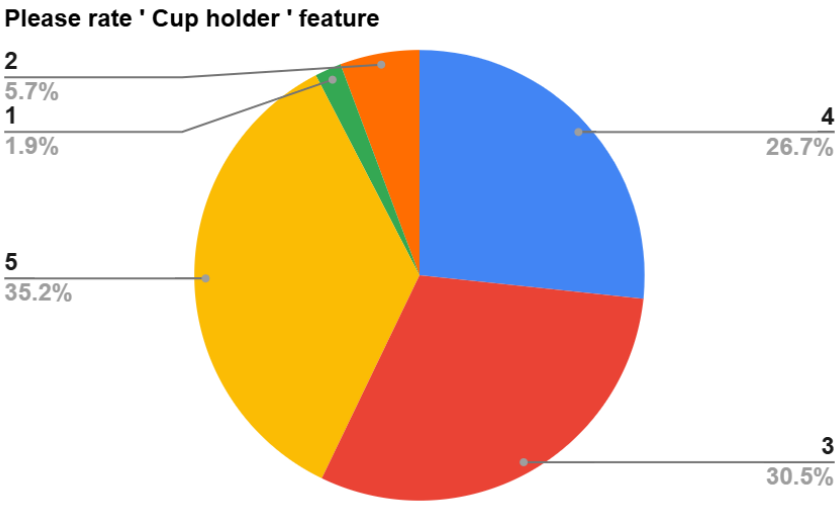


Figure 5.22 Respondents about the features of cup holder

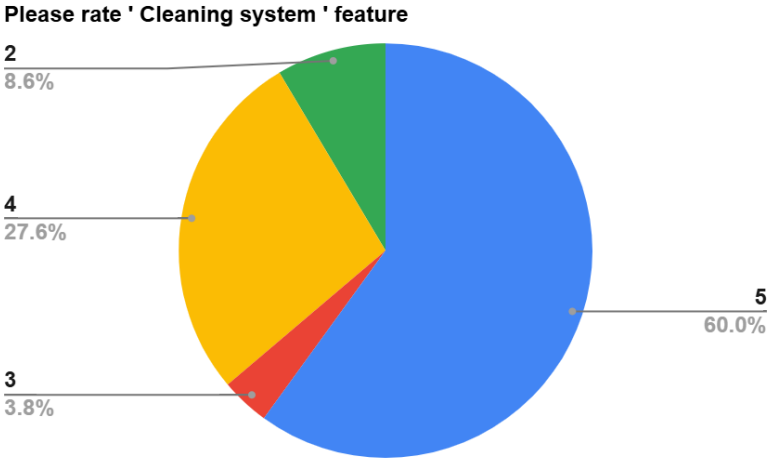


Figure 5.23 Respondents about the features of cleaning system

Insight: Fig 3.18-3.22 dictates that customers are positive regarding our integrated feature at wheelchairs. The feedback gathered through these figures highlights a strong acceptance and approval of the innovative elements we have incorporated, such as the convertible bed function and self-cleaning technology.

Chapter 04: Kano Model Analysis

4.1 Kano Model Needs Assessment

Customer preferences are divided into five categories by the Kano Model, a theory for product creation and customer satisfaction: Must be Quality, One-dimensional Quality, Attractive Quality, Indifferent Quality, as well as Reverse Quality.

4.2 Evaluation of Customer Needs

	Dysfunctional					
Functional		Like	Must-be	Neutral	Live-with	Dislike
	Like	Q	A	A	A	O
	Must-be	R	I	I	I	M
	Neutral	R	I	I	I	M
	Live-with	R	I	I	I	M
	Dislike	R	R	R	R	Q

Table 6.1 Kano Evaluation table

1. Must-be Quality (Basic Needs)
Customers take these things for granted and anticipate them. In the event that these requirements are not met, clients will be very unhappy.

2. One-dimensional Quality (Performance Needs)
These are characteristics that, when met, lead to satisfaction; when unmet, they lead to discontent. Customers specifically request these features.

3. Attractive Quality (Excitement Needs)
These are characteristics that surprise and thrill clients when they are offered. Their presence greatly raises contentment, while their absence does not lead to discontent.

4. Indifferent Quality
Customers are not concerned with these features. The satisfaction of customers is unaffected by their presence or disappearance.

5. Reverse Quality
Some customers prefer these features, but others don't. Some people may be satisfied by their existence, while others may not be.

4.3 Analysis of Kano Model

We did the kano model customer need analysis which is a tool that differentiates potential features according to users' needs. Based on our survey, we took our 4 popular features and asked 100+ qualified respondents about a pair of functional and dysfunctional perception questions for each feature.

To better comprehend the needs and pleasures of the users, we conducted both discrete and continuous analyses.

4.3.1 Continuous Analysis

The respondents' functional and dysfunctional scores for comparing the features are averaged in four quadrants, Must-be, One dimensional, Attractive, and Indifferent. Fig. 2 shows the result of the continuous analysis.

Continuous Analysis			
Feature Name	**Dysfunctional (X)**	**Functional (Y)**	**Category**
180 Degree Decline	2.607142857	1.982142857	Must Be
Defecation System	2.508928571	1.803571429	Must Be
Leg, Arm and Head rest	2.598214286	1.955357143	Must Be
Cup Holder	1.705357143	2.241071429	Indifferent
Cleaning system	2.410714286	1.973214286	Must Be

Table 6.2 Continuous Analysis

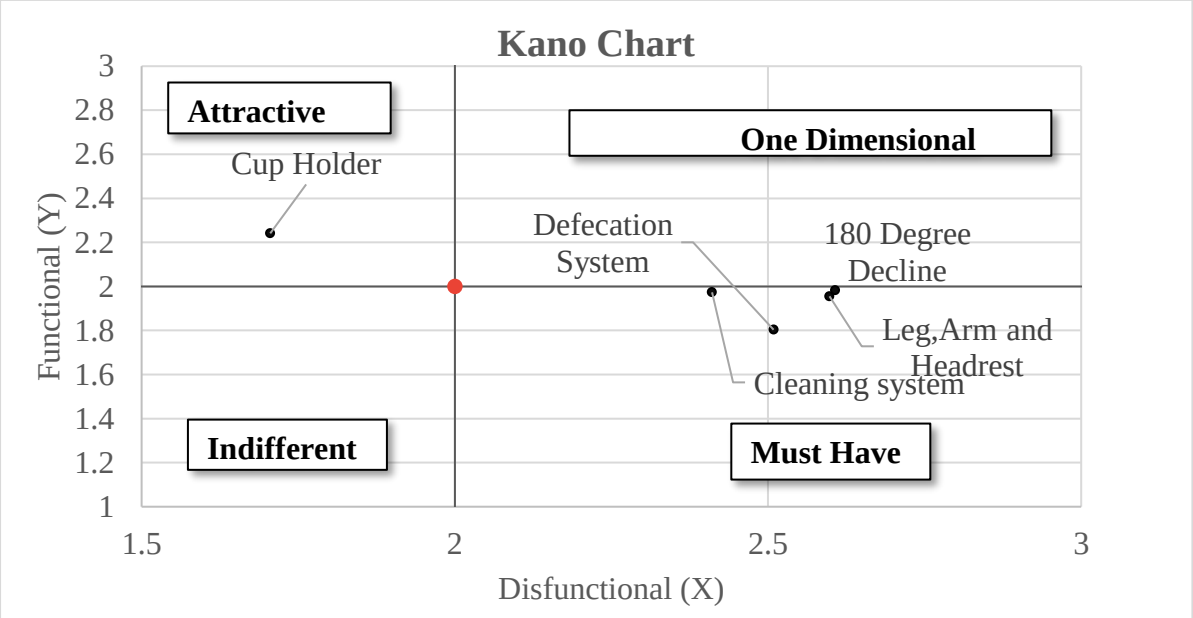


Figure 6.1 Continuous analysis of Kano model

The chart indicates that the most critical features in the wheelchair bed (those placed in the One Dimensional and Must Have categories) include the Defecation System, 180 Degree Decline, Leg, Arm, and Headrest, and Cleaning System. These features are essential for user satisfaction and should be prioritized in design. In contrast, while features like a Cup Holder can enhance the product's appeal, they are not as critical and can be considered secondary additions.

4.3.2 Discrete Analysis

The customers' answers were projected and evaluated for every feature of our product. In table 1, the features are categorized to represent the largest proportion of the respondents. Discrete Analysis is the preferred way of looking at Kano results as the cut-offs and numerical weights of different categories are not calibratable for each study in continuous analysis.

Discrete Analysis							
Feature Name	**M**	**O**	**A**	**I**	**R**	**Q**	**Evaluation**
180 Degree Decline	45.54%	14.29%	8.04%	26.79%	3.57%	0.00%	Must Be
Defecation System	43.75%	11.61%	8.04%	30.36%	5.36%	0.00%	Must Be
Leg, Arm and Headrest	39.29%	14.29%	8.93%	34.82%	0.89%	0.00%	Must Be
Cup Holder	17.86%	15.18%	29.46%	30.36%	4.46%	0.00%	Indifferent
Cleaning system	40.18%	9.82%	13.39%	32.14%	3.57%	0.00%	Must Be

Table 6.3 Findings of discrete analysis

5.4 Our Findings from the Kano Model

Feature No	**Feature Name**	**Customer's Requirement**
01	180 Degree Decline	Must Be
02	Defecation System	Must Be
03	Leg, Arm and Headrest	Must Be
04	Cup Holder	Indifferent
05	Cleaning System	Must Be

Table 6.4 . Findings of Kano Model

Chapter 05: Competitor Analysis

Competitor analysis for launching a convertible wheelchair involves studying existing wheelchair brands and their product features. This analysis reveals market trends and customer needs, guiding to incorporate innovative features that set your product apart. Understanding competitors' pricing and marketing strategies also helps in positioning your product more effectively. Ultimately, competitor analysis ensures your convertible wheelchair meets market demands, offering superior value and gaining a competitive edge.

5.1 Comparing Our Features with the Market Competitors

Competitor Feature Comparison Card		My Product	Competitor A (Smart Foldable Electric wheelchair)	Competitor B (Manual Folding Wheelchair)	Competitor C (Commode Wheelchair)	Competitor D (Sleeping wheelchair)
Attribute #1	180 degree Decline	✓	X	X	X	✓
Attribute #2	Defecation system	✓	X	X	✓	X
Attribute #3	Leg,arm & head rest	✓	✓	✓	✓	✓
Attribute #4	Cup Holder	✓	✓	X	X	X
Attribute #5	Self Claening System	✓	X	X	X	X
Attribute #6	360 degree rotation of chair	X	✓	X	X	X

Figure 7.1 Market Competitors

5.2 SWOT Analysis

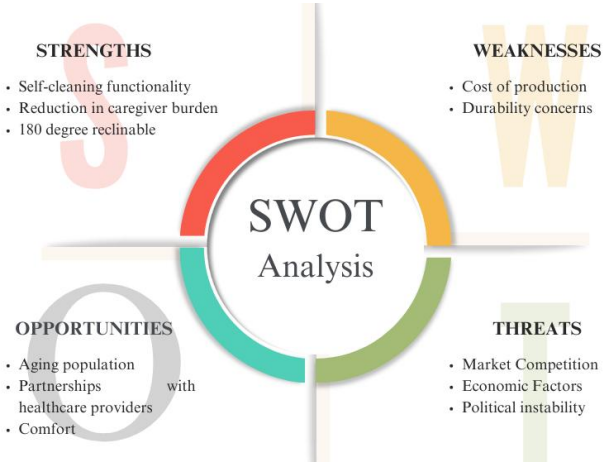


Figure 7.2 SWOT Analysis

5.3 Competitor Categorical Analysis

Competitor Categorical Analysis						
		My Product	Competitor A (Smart Foldable wheelchair)	Competitor B (Manual Folding Wheelchair)	Competitor C (Commode Wheelchair)	Competitor D (Sleeping Wheelchair)
Product Highlights						
Product Profile	1. Conversion to wheelchair to bed 2.Defecation System 3. Self cleaning 4.Comfort 5.Lightweight	4	3	1	2	2
Key Competitive Advantage	1)Durability 2)Lightweight3)Easily Convertible 4)Defecation Facility	5	3	2	3	2
Total		9	6	3	5	4
Product Information						
180 degree decline		5	1	1	1	5
Defecation System		5	1	1	5	1
Leg,arm & Headrest		5	5	3	3	3
Cup Holder		4	5	2	1	1
Cleaning System		5	1	2	2	1
Price		4	3	4	3	4
Total		28	16	13	15	24
Overall Total		37	22	16	20	28

Figure 7.3 Competitor Categorical Analysis

Chapter 06: Functional Structure Development

6.1 Introduction

Developing a functional structure in product design is essential to ensure that the design process is organized, efficient, and focused on achieving the product's intended functions and objectives. A functional structure breaks down the design into logical components or functions, defining the relationships between them.

Steps of functional structure development include:

1. Introduce the function of the product
2. State the sub-function
3. Order the sub-function
4. Define the sub-function.

6.2 Blackbox

In product design, a "black box" refers to a component or system whose internal workings are not transparent or visible to the user or the outside observer. It means that the user knows what input to provide and what output to expect, but the specific processes that occur inside the black box are hidden or abstracted from the user's view. This concept is often used in various fields, including engineering, software development, and electronics. The input-output relationship of the convertible wheelchair bed is represented by a Blackbox:

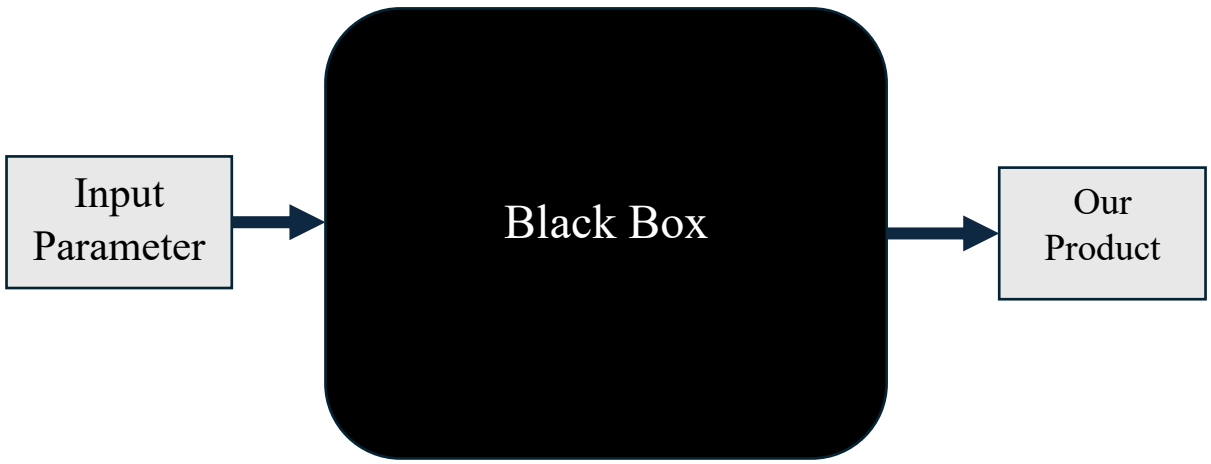


Figure 8.1 Black Box diagram of Convertible Wheelchair-Bed

6.3 Functional Tree

A Functional Tree is a systematic representation of the functions required to achieve a particular objective or to fulfill the design of a product, system, or process. It provides a hierarchical breakdown of the main function and its sub-functions, illustrating the relationship between them. This tool is commonly used in engineering design, product development, and systems analysis to ensure that every aspect of the system's operation is considered and accounted for.

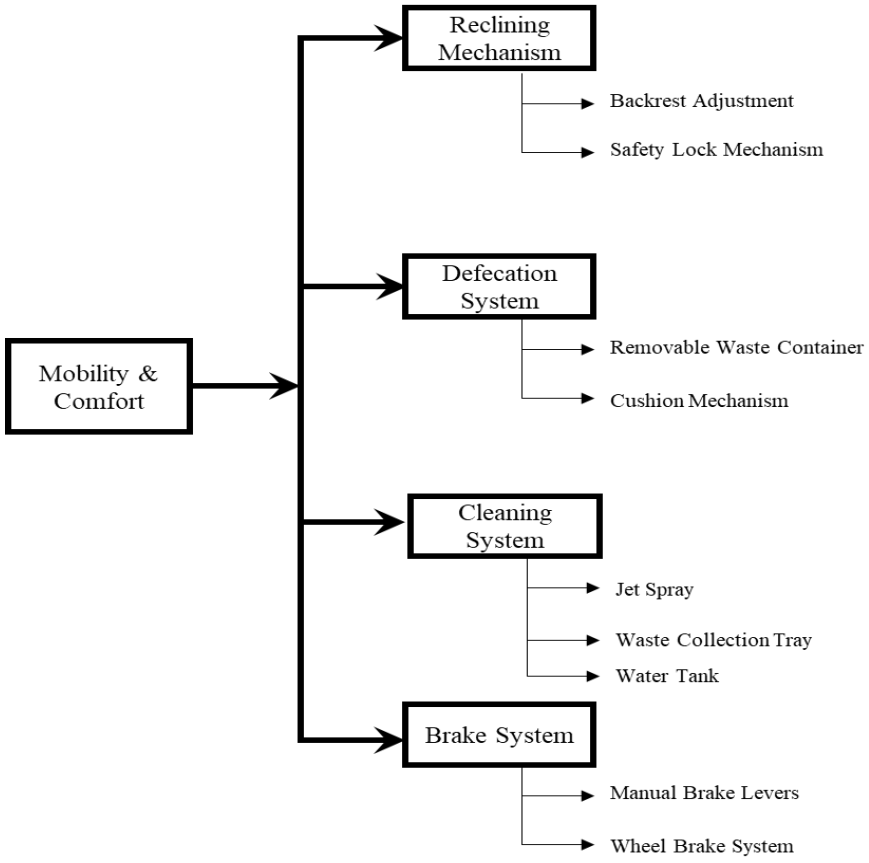


Figure 8.2 Functional Tree of Convertible Wheelchair-Bed

6.4 Functional Structure

A Functional Structure is a hierarchical framework used to organize a system, product, or process based on its individual functions or activities. It breaks down a complex system into smaller, manageable components, each responsible for a specific function that contributes to the overall objective. This structure helps in clarifying roles, responsibilities, and relationships within a system, making it easier to design, develop and manage.

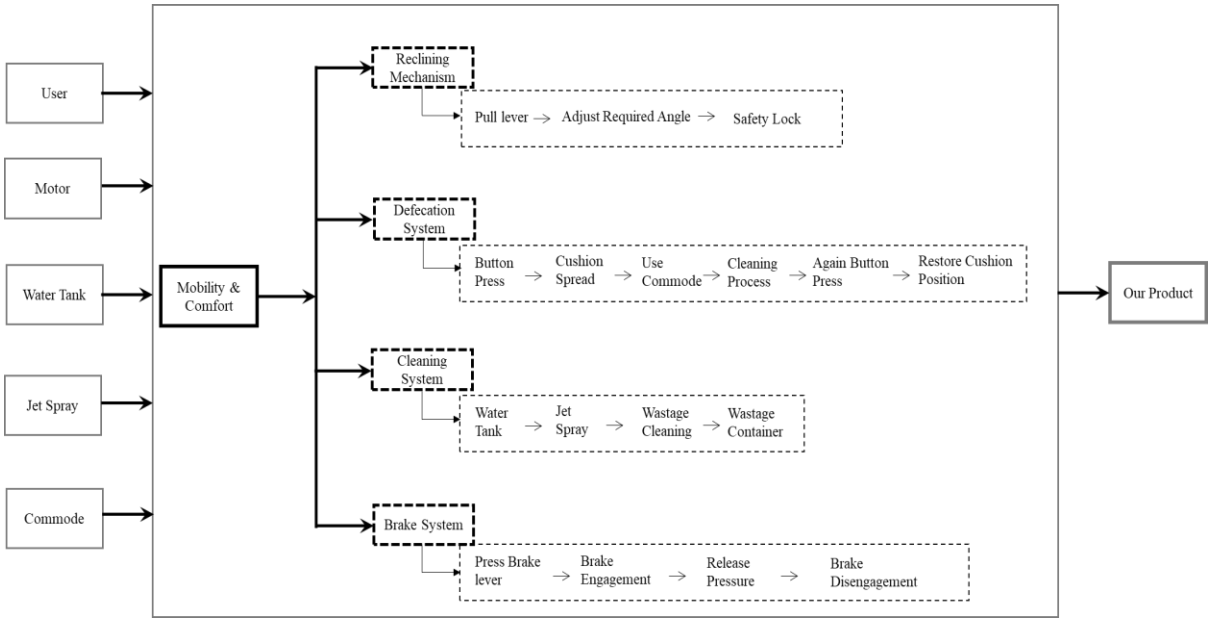


Figure 8.3 Functional Structure diagram of Convertible Wheelchair-Bed

Chapter 07: Quality Function Development

7.1 Objective of QFD

QFD is a systematic process for identifying customers' needs so that a product can be designed to meet their demands. The QFD method employs a matrix format that resembles a home, it is also known as the “House of Quality” (HoQ). The strategy assists in making informed decisions about customer requirements. QFD aims to convert subjective and even objective quality requirements into quantifiable and measurable quality standards. It is a complementary approach for indicating how and where priorities should be assigned in product development**.**

8.2 House of Quality

Type of correlation	Symbol
Strongly Positive	+
No correlation	
Strongly Negative	-

Table 9.1 Symbol to illustrate correlations between technical requirements

Types of relation	Symbol	Value
No Relation		**0**
Weak Relation	**∇**	**1**
Moderate Relation	**o**	**3**
Strong Relation	**•**	**9**

Table 9.2 Symbol to illustrate correlations between technical requirements

Improvement	Symbol
Maximize	▲
Minimize	▼
Target	◇

Table 9.3 Direction of improvement for functional requirements

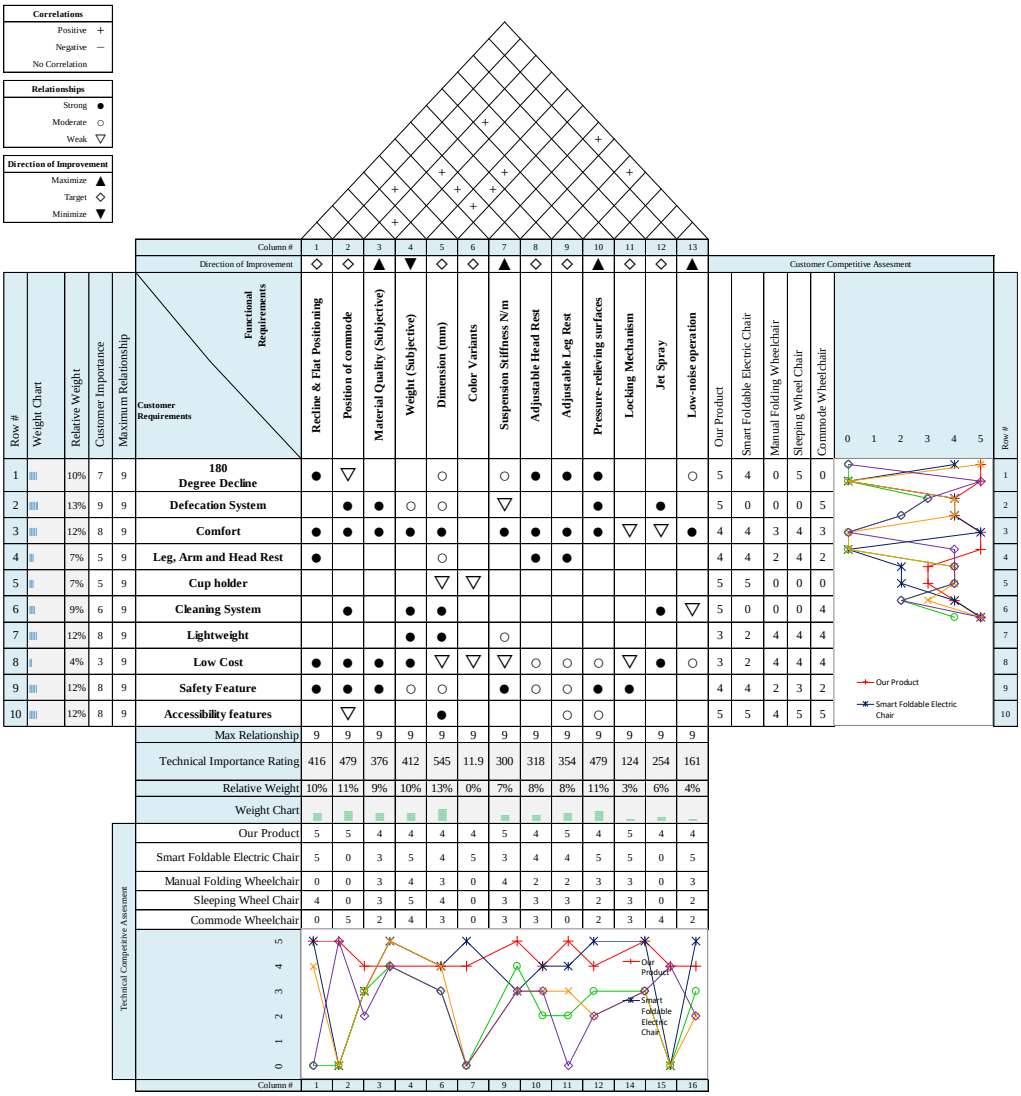


Figure 9.1 House of Quality

8.3 Finding of QFD

Features Name	Technical Specifications	Direction of Improvement
Cost	12-14K BDT	Minimize
Recline & Flat positioning	Moveable and 180-degree decline	Target
Defecation system	Manually removable waste container integrated with a cushion-activated defecation system	Target
Breaking Mechanism	Manual brake levers for user control and wheel brake levers for enhanced stability and safety	Target
Material Quality	Lighter material with enough durability should be used for less weight and more strength.	Maximize

Table 9.4 Finding of QFD

Chapter 08: Design and Specification Analysis

A product specification is a document with a requirement that provides all sorts of information that is necessary to provide dimensions, specifications, and requirements to build all the functions of the product. A good product is one that is functional, reliable, and effective, and that is designed to be economical and aesthetically pleasing. The product specification aims to provide all necessary details to build the product in the most cost-effective manner to satisfy the customer's need.

8.1 Specification Chart:

Specifications	Value
Height	1100 - 1200 mm
Length	900-1000 mm
Weidth	400 - 500 mm
Maximum weight	120 kg
Waste Container	Water resistants
Large Wheels	556 mm Diameter
Small Wheels	140 mm Diameter
Manual Pump	1 Piece
Water Container	3.5 liter

Table 10.1 Specification chart

8.2 3D View of wheelchair:

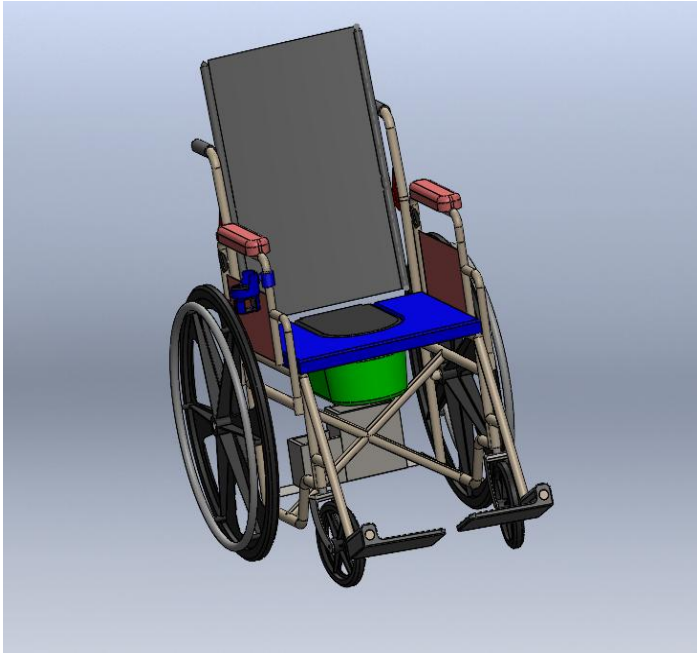


Figure 10.1 3D model of wheelchair

8.3 2D View of wheelchair:

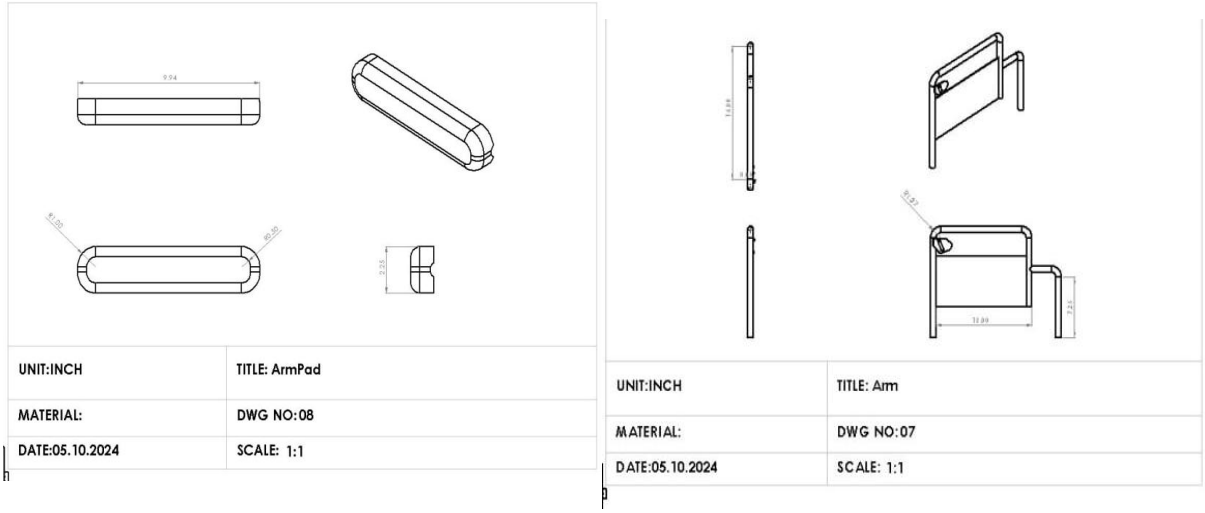


Figure 10.2 2D view and Different side view of Armpad & Arm

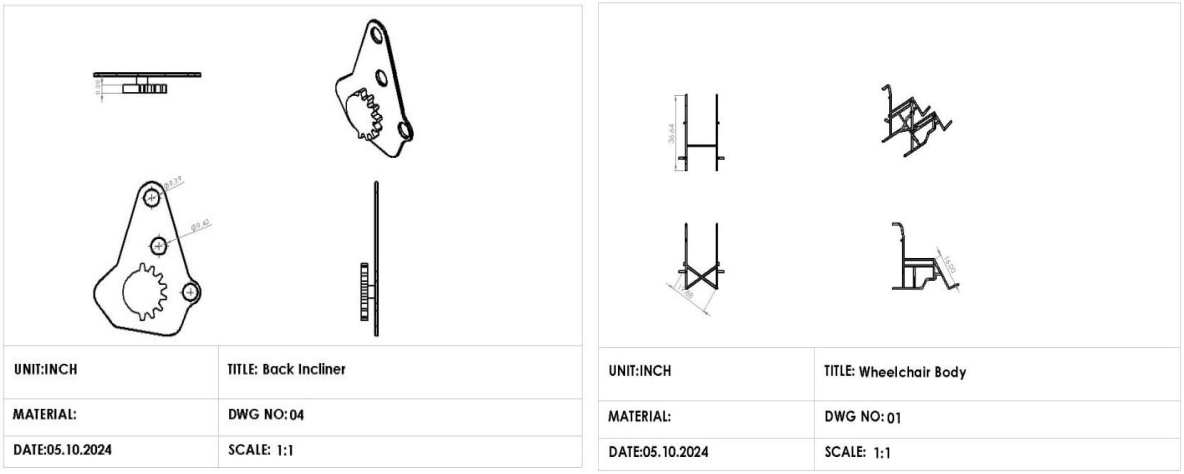
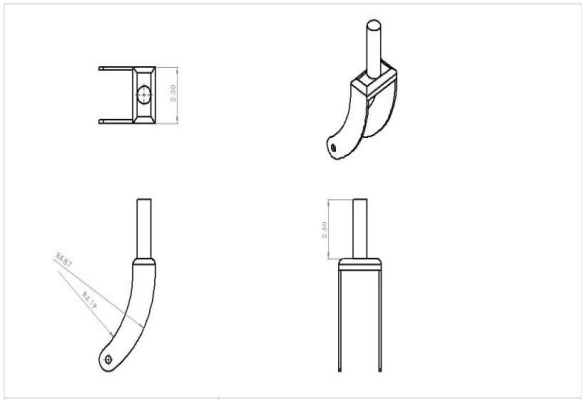
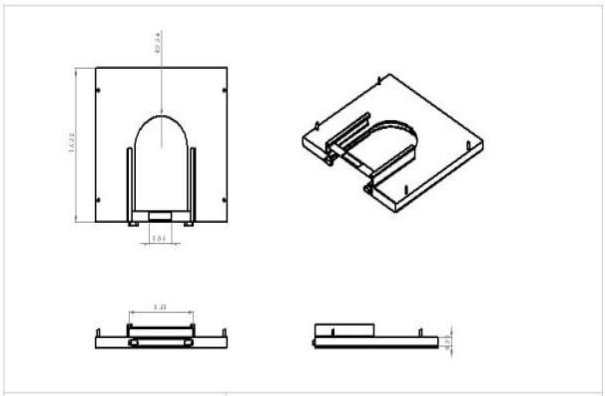


Figure 10.3 2D view and Different side view of Back Incliner & Body

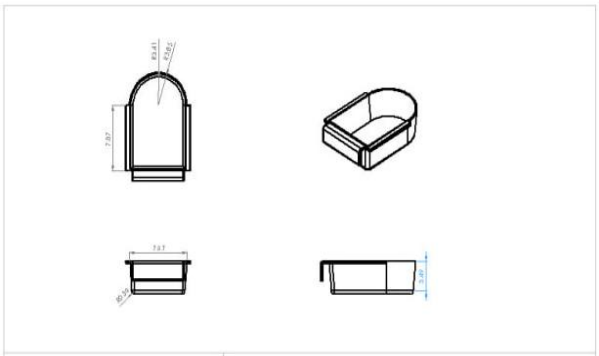


UNIT:INCH	TITLE: CastleFork
MATERIAL:	DWG NO:09
DATE:05.10.2024	SCALE: 1:1

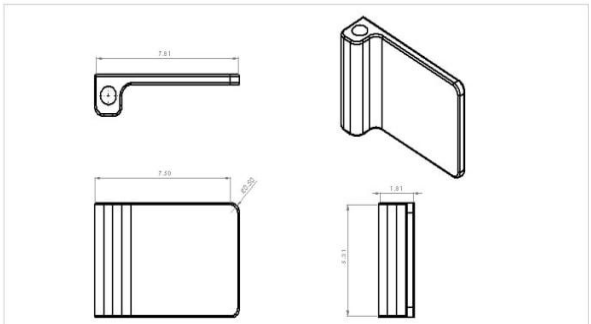


UNIT:INCH	TITLE: Commode Seat
MATERIAL:	DWG NO: 02
DATE:05.10.2024	SCALE: 1:1

Figure 10.4 2D view and Different side view of CastleFork & Commode seat

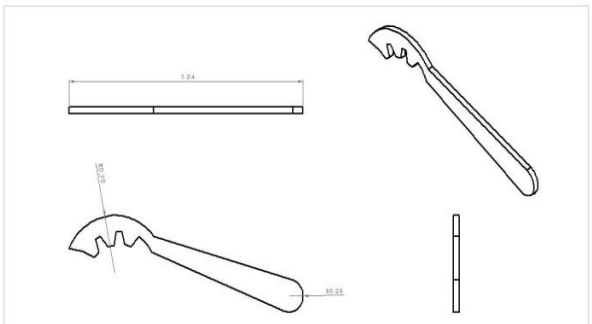


UNIT:INCH	TITLE: Waste Container
MATERIAL:	DWG NO: 03
DATE:05.10.2024	SCALE: 1:1

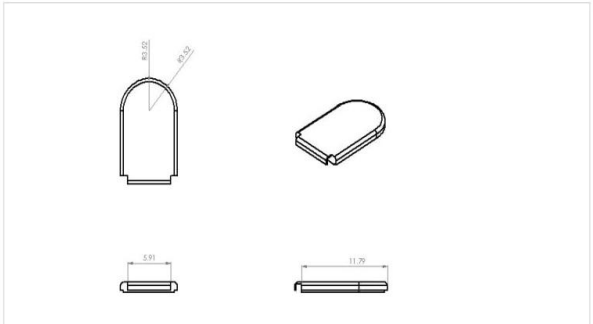


UNIT:INCH	TITLE: Foot Rest
MATERIAL:	DWG NO:10
DATE:05.10.2024	SCALE: 1:1

Figure 10.5 2D view and Different side views of Waste Container & Foot Rest

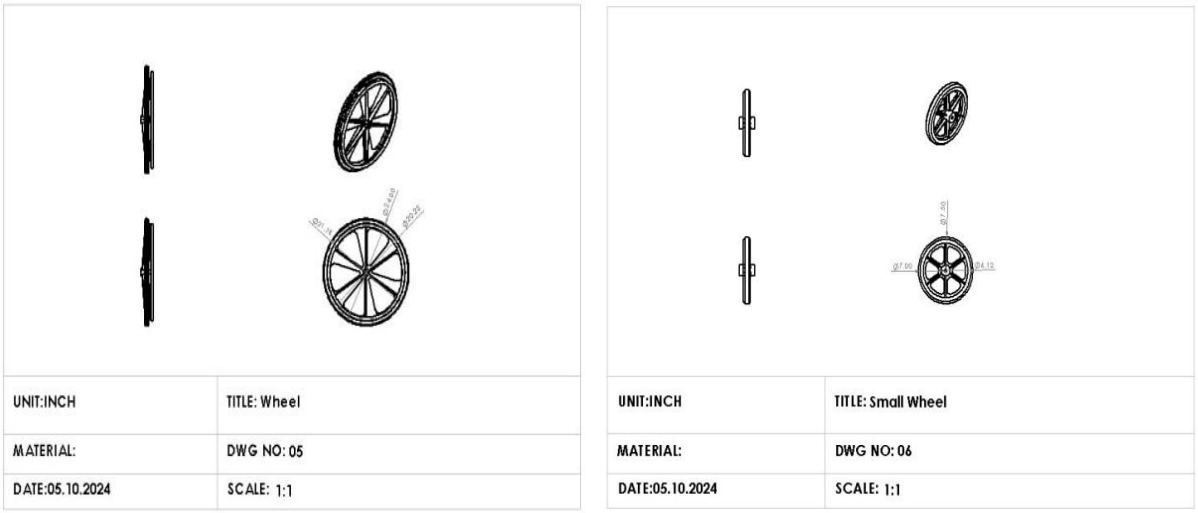


UNIT:INCH	TITLE: Front Incliner
MATERIAL:	DWG NO:11
DATE:05.10.2024	SCALE: 1:1



UNIT:INCH	TITLE: Seat Top
MATERIAL:	DWG NO: 13
DATE:05.10.2024	SCALE: 1:1

Figure 10.6 2D view and Different side view of Foot Incliner & Seat top



UNIT:INCH	TITLE: Wheel
MATERIAL:	DWG NO: 05
DATE:05.10.2024	SCALE: 1:1

UNIT:INCH	TITLE: Small Wheel
MATERIAL:	DWG NO: 06
DATE:05.10.2024	SCALE: 1:1

Figure 10.7 3D view and Different side view of wheel

Chapter 09: Stress Analysis

Stress analysis is a fundamental aspect of engineering mechanics that involves the study and evaluation of forces, pressures, and deformations acting on materials and structures. It plays a critical role in ensuring the safety, functionality, and durability of components under various loading conditions.

The primary goal of stress analysis is to determine the stresses and strains that develop within a material or structure when subjected to external forces, thermal effects, or environmental conditions. By understanding these internal responses, engineers can predict the likelihood of material failure, deformation, or instability, enabling the design of efficient and reliable systems.

In this analysis we have analyzed stress, strain, deformation and factor of safety of Wheelchair different part. To conduct this analysis, we have used Solidworks software.

9.1 Stress Analysis of Wheelchair Frame:

Study Properties

Study name	Static 2
Analysis type	Static
Mesh type	Solid Mesh
Thermal Effect:	On
Thermal option	Include temperature loads
Zero strain temperature	298 Kelvin
Include fluid pressure effects from SOLIDWORKS Flow Simulation	Off
Solver type	Automatic
Inplane Effect:	Off
Soft Spring:	Off
Inertial Relief:	Off
Incompatible bonding options	Automatic
Large displacement	Off
Compute free body forces	On
Friction	Off
Use Adaptive Method:	Off
Result folder	SOLIDWORKS document (E:)

Table 11.1 Study Properties

Material Properties

Model Reference	Properties		Components
	Name:	**6061-O (SS)**	**SolidBody 1(Revolve1)(wheel chair -frame)**
	Model type:	**Linear Elastic Isotropic**	
	Default failure criterion:	**Max von Mises Stress**	
	Yield strength:	**6.20528e+07 N/m^2**	
	Tensile strength:	**1.25e+08 N/m^2**	
	Elastic modulus:	**6.9e+10 N/m^2**	
	Poisson's ratio:	**0.33**	
	Mass density:	**2,700 kg/m^3**	
	Shear modulus:	**2.6e+10 N/m^2**	
	Thermal expansion coefficient:	**2.4e-05 /Kelvin**	

Table 11.2 Material Properties

Study Results

Name	Type	Min	Max
Stress1	VON: von Mises Stress	3.033e+00N/m^2	1.25e+08N/m^2

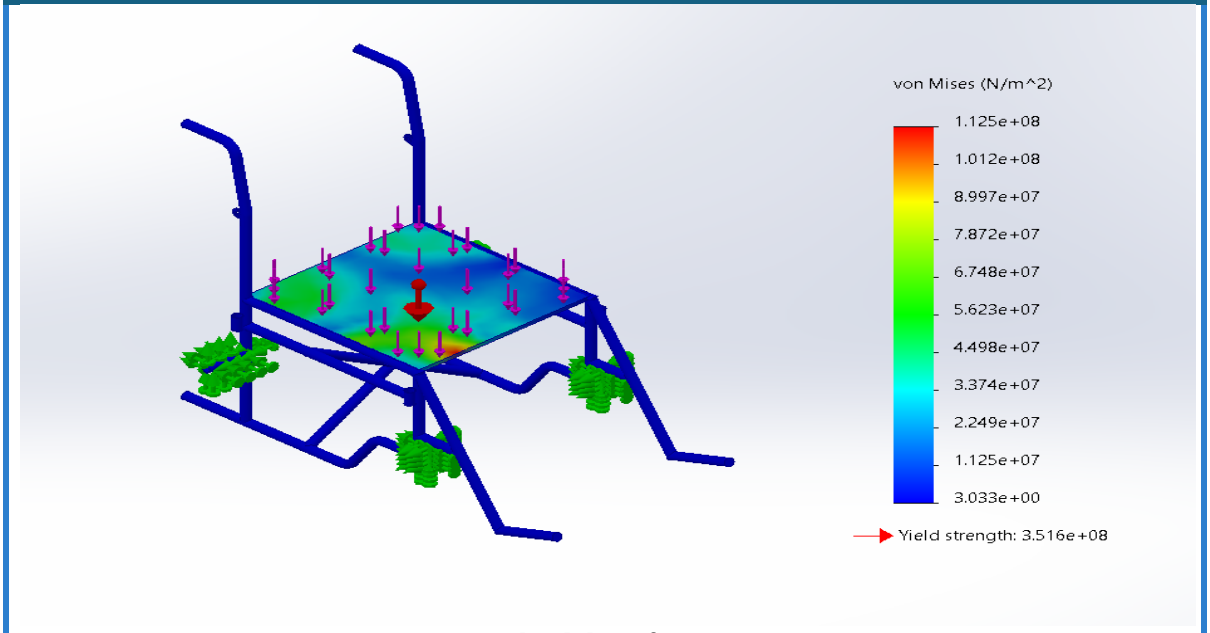
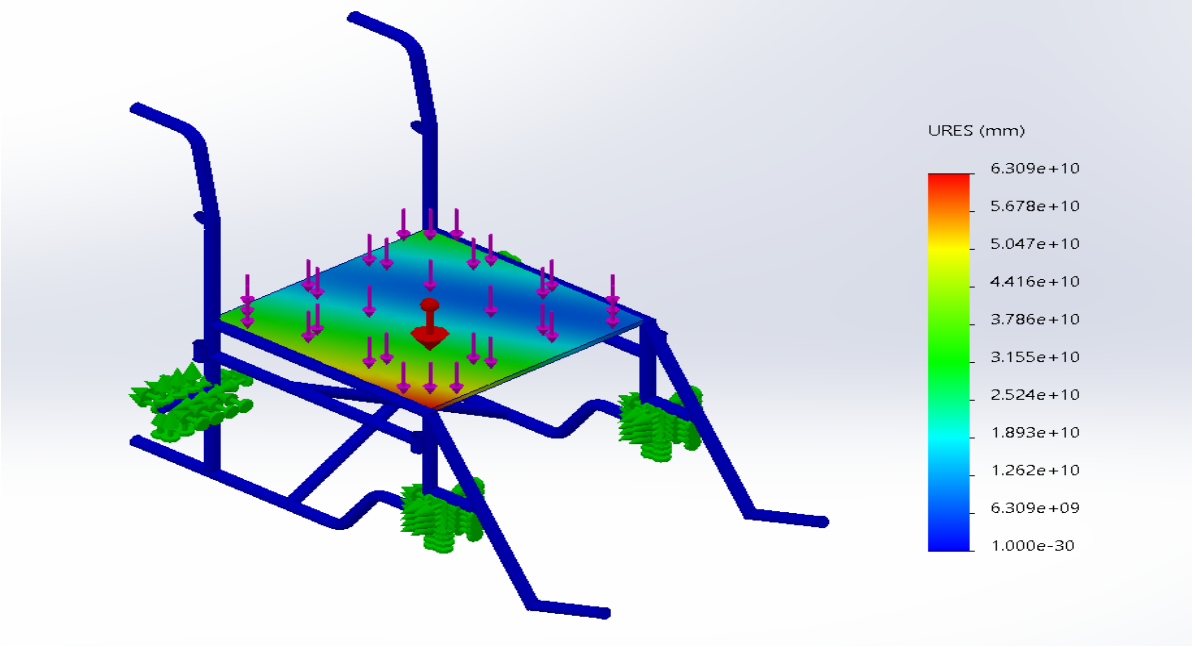
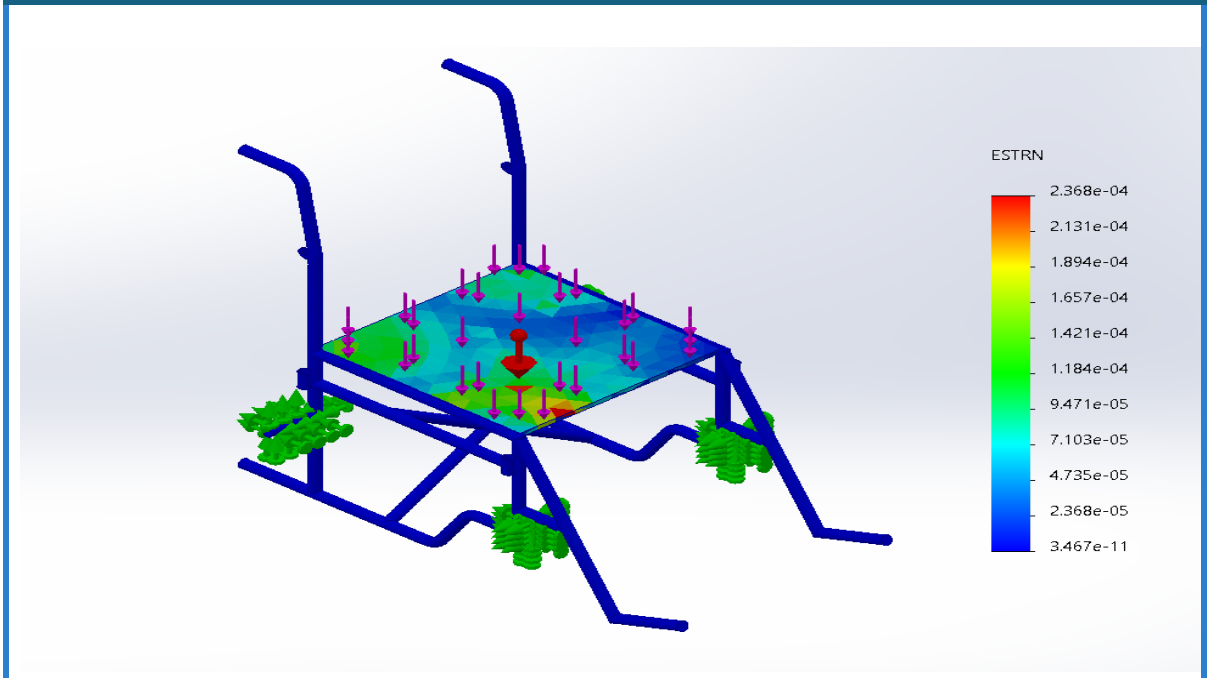
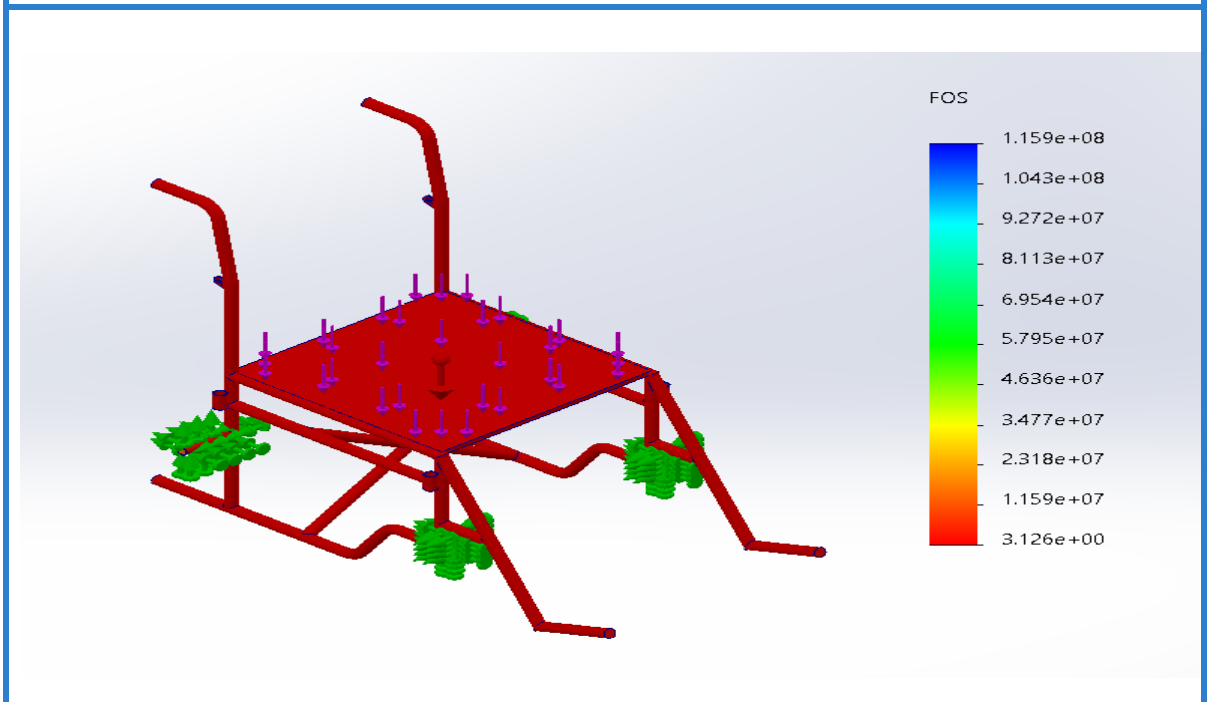
Name	Type	Min	Max
			
Wheelchair frame			

Table 11.3 Study Results

Name	Type	Min	Max
Displacement1	URES: Resultant Displacement	0.000e+00mm Node: 41	3.870e-06mm Node: 136116



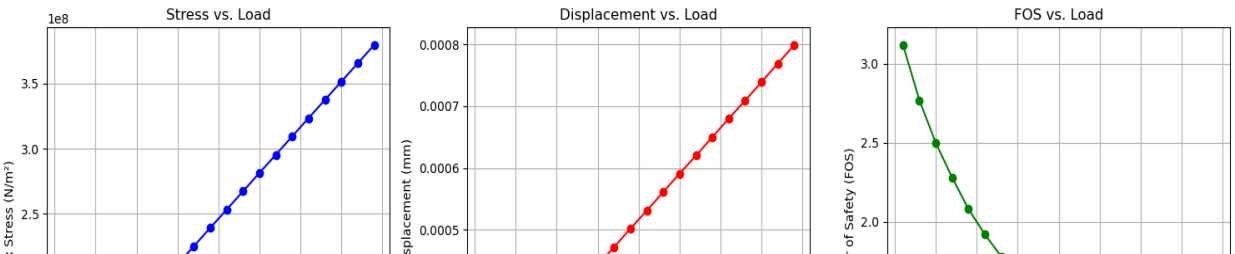
Wheelchair frame			
Name	**Type**	**Min**	**Max**
Strain1	ESTRN: Equivalent Strain	3.467e+-11	2.368e-04

Name	Type	Min	Max
			
			
Name	Type	Min	Max
FOS1	FOS: Factor of Safety	3.126e+00	1.159e+08

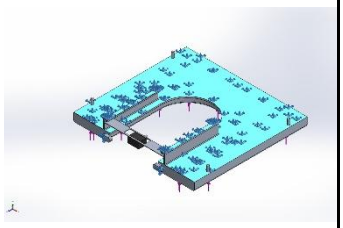
9.2 Effect of varying Load (800N-2700N)

Load (N)	Max Stress (Pa)	Max Displacement (m)	Factor of Safety (FOS)
800	1.125×10^8	0.000237	3.12
900	1.266×10^8	0.000267	2.77
1000	1.406×10^8	0.000296	2.5
1100	1.547×10^8	0.000325	2.28
1200	1.688×10^8	0.000355	2.08
1300	1.828×10^8	0.000384	1.92
1400	1.969×10^8	0.000414	1.78
1500	2.109×10^8	0.000443	1.68
1600	2.250×10^8	0.000472	1.56
1700	2.391×10^8	0.000502	1.47
1800	2.531×10^8	0.000531	1.39
1900	2.672×10^8	0.000561	1.32
2000	2.813×10^8	0.000591	1.25
2100	2.953×10^8	0.00062	1.19
2200	3.094×10^8	0.00065	1.14
2300	3.235×10^8	0.00068	1.09
2400	3.375×10^8	0.000709	1.04
2500	3.516×10^8	0.000739	1
2600	3.656×10^8	0.000769	0.96
2700	3.797×10^8	0.000799	0.92

Table 11.4 Effect of Varing Load



Loads and Fixtures

Fixture name	Fixture Image	Fixture Details
Fixed-1		**Entities:** 5 face(s) **Type:** Fixed Geometry

Resultant Forces

Components	X	Y	Z	Resultant
Reaction force(N)	-6.44112e-05	-899.997	0.000165796	899.997
Reaction Moment(N.m)	0	0	0	0

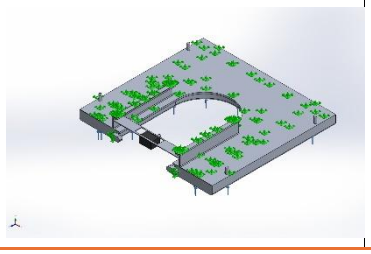
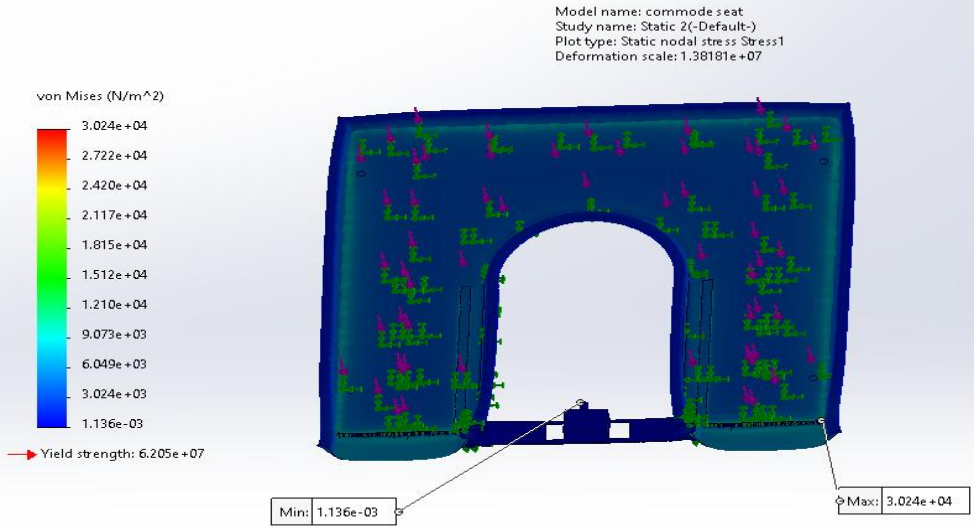
Load name	Load Image	Load Details
Force-1		**Entities:** 1 face(s) **Type:** Apply normal force **Value:** 900 N

Figure 11.1 Effect of Varing load

9.3 Stress analysis of Commode Seat

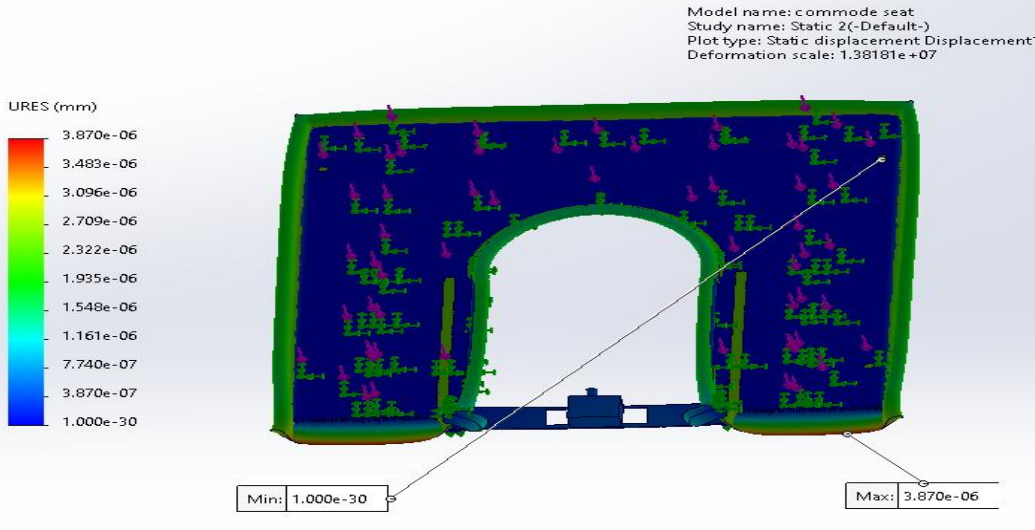
Study Results

Name	Type	Min	Max
Stress1	VON: von Mises Stress	1.136e-03N/m^2 Node: 63475	3.024e+04N/m^2 Node: 83



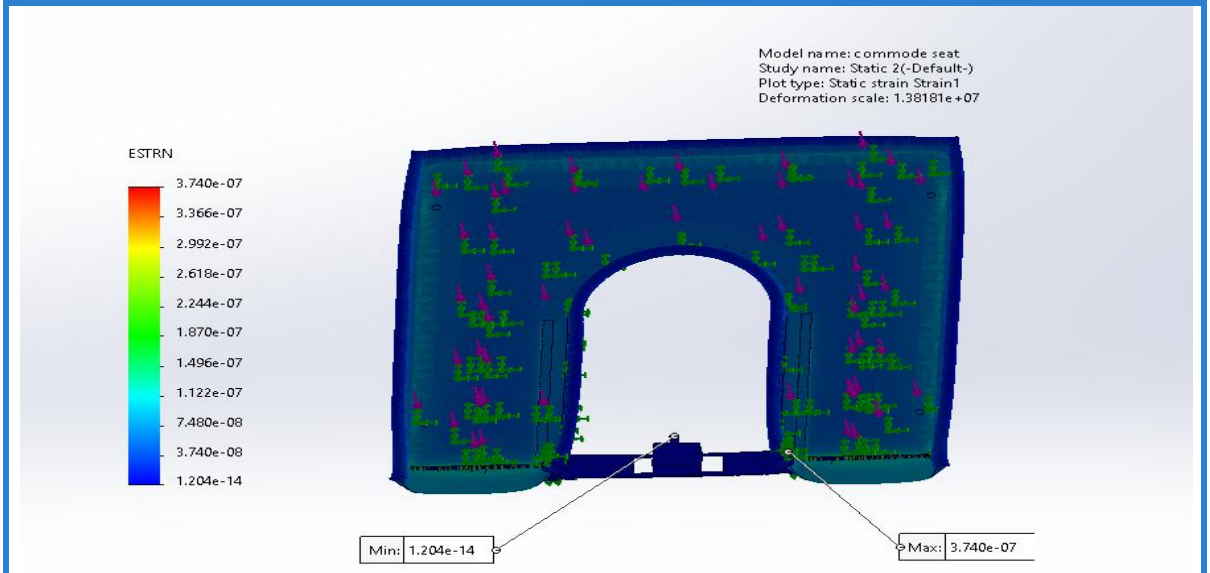
commode seat-Static 2-Stress-Stress1

Name	Type	Min	Max
Displacement1	URES: Resultant Displacement	0.000e+00mm Node: 41	3.870e-06mm Node: 136116



commode seat-Static 2-Displacement-Displacement1

Name	Type	Min	Max
Strain1	ESTRN: Equivalent Strain	1.204e-14 Element: 31113	3.740e-07 Element: 82768

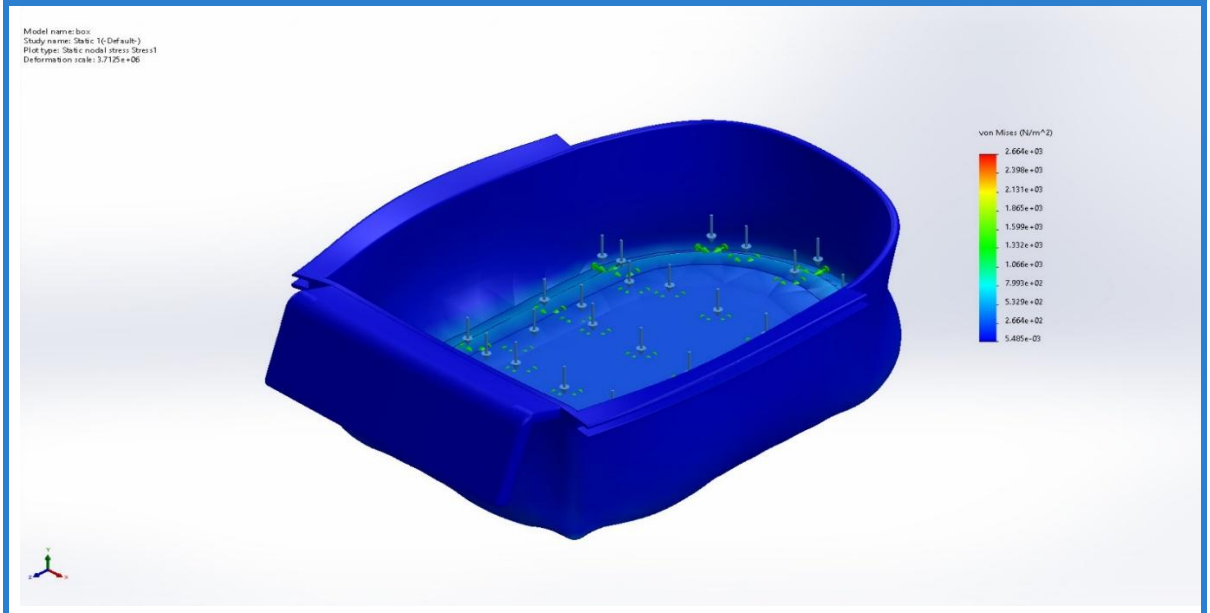


commode seat-Static 2-Strain-Strain1

9.4 Stress Analysis of Commode

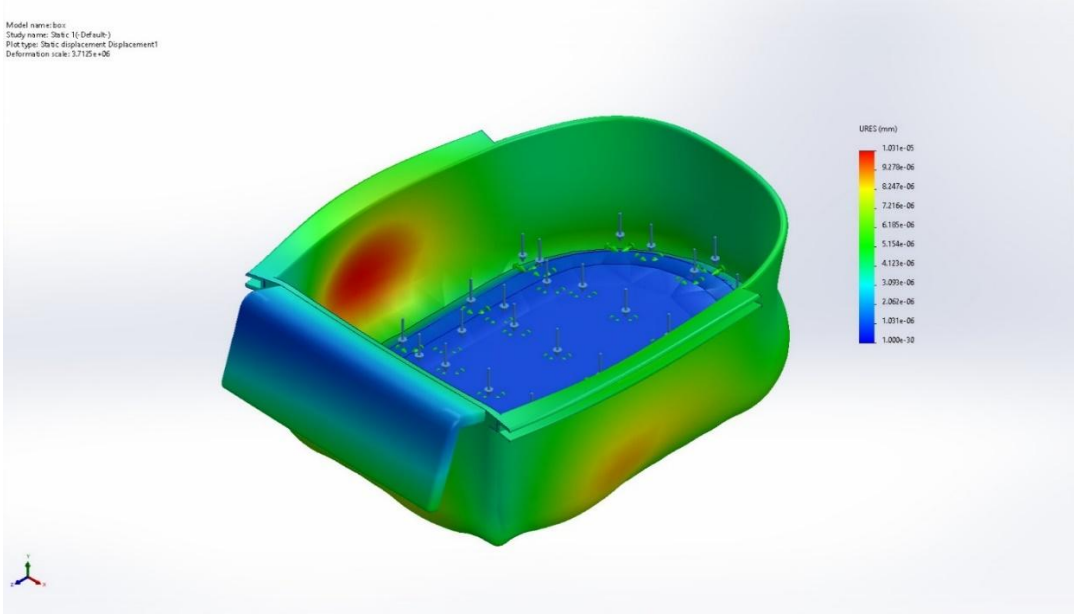
Study Results

Name	Type	Min	Max
Stress1	VON: von Mises Stress	5.485e-03N/m^2 Node: 149245	2.664e+03N/m^2 Node: 118652



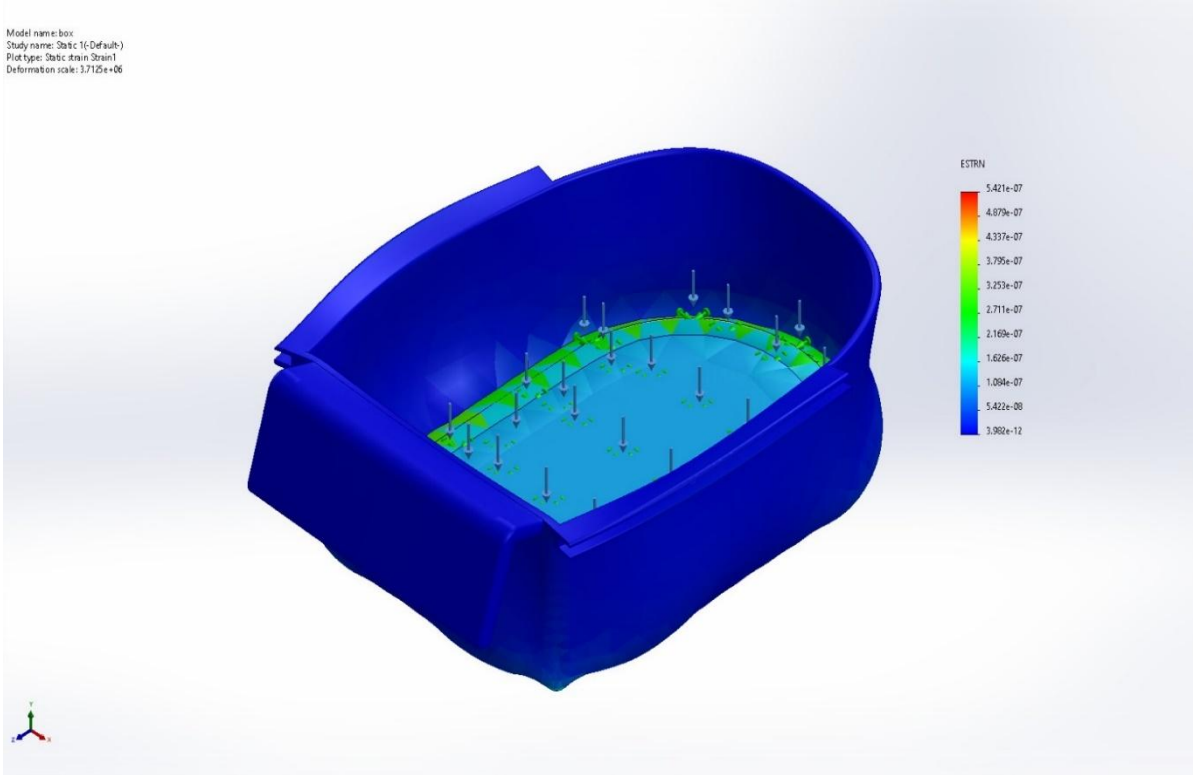
box-Static 1-Stress-Stress1

Name	Type	Min	Max
Name	Type	Min	Max
Displacement1	URES: Resultant Displacement	0.000e+00mm Node: 30	1.031e-05mm Node: 4562



box-Static 1-Displacement-Displacement1

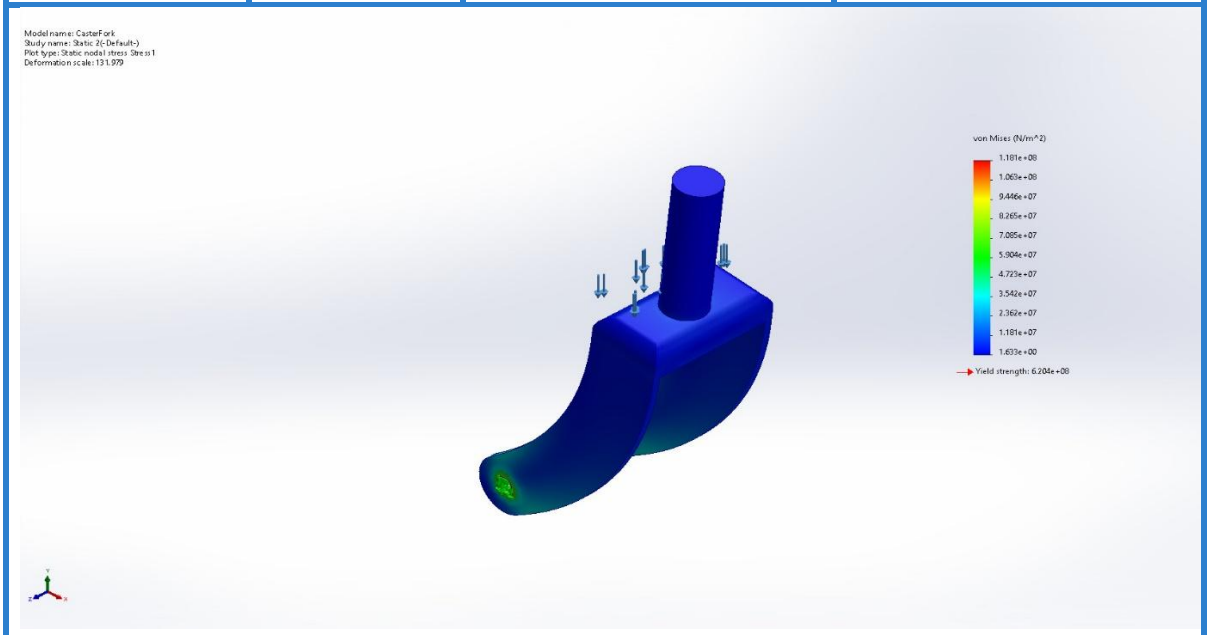
Name	Type	Min	Max
Strain1	ESTRN: Equivalent Strain	3.982e-12 Element: 86097	5.421e-07 Element: 63379



9.5 Stress analysis of Castle fork

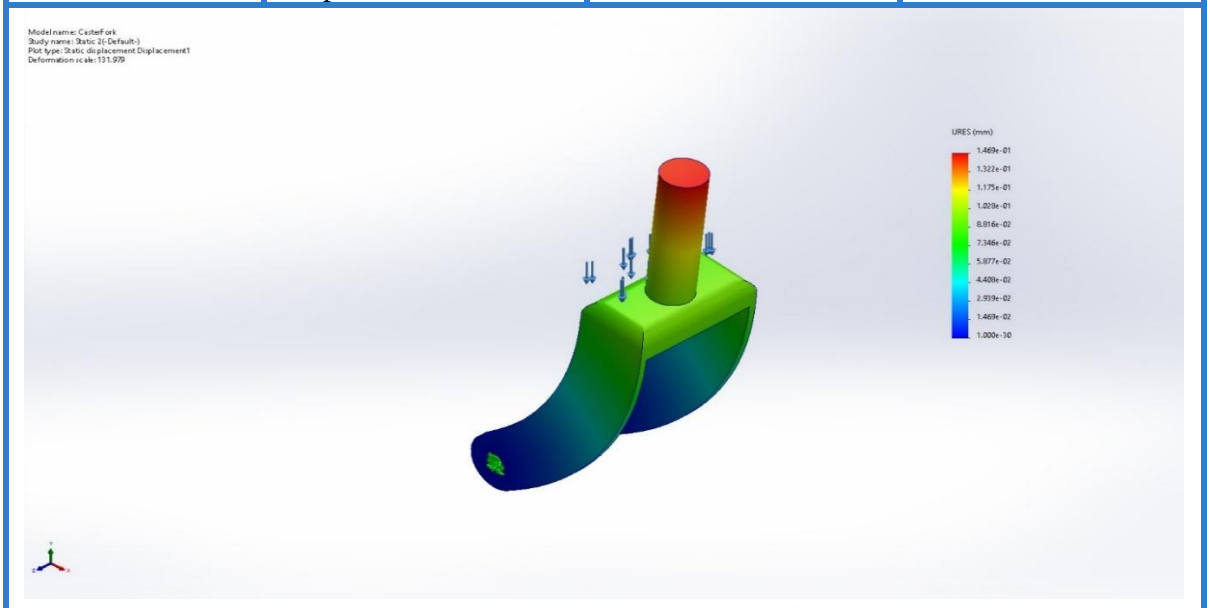
Study Results

Name	Type	Min	Max
Stress1	VON: von Mises Stress	1.633e+00N/m^2 Node: 8725	1.181e+08N/m^2 Node: 13000



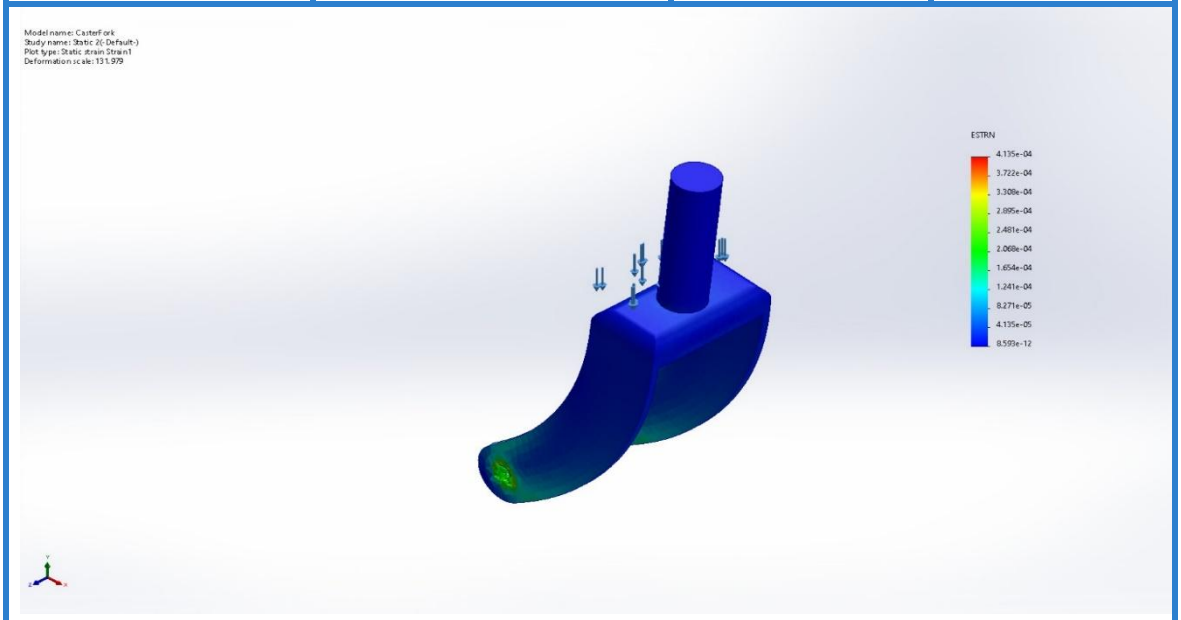
CasterFork-Static 2-Stress-Stress1

Name	Type	Min	Max
Displacement1	URES: Resultant Displacement	0.000e+00mm Node: 18	1.469e-01mm Node: 1585



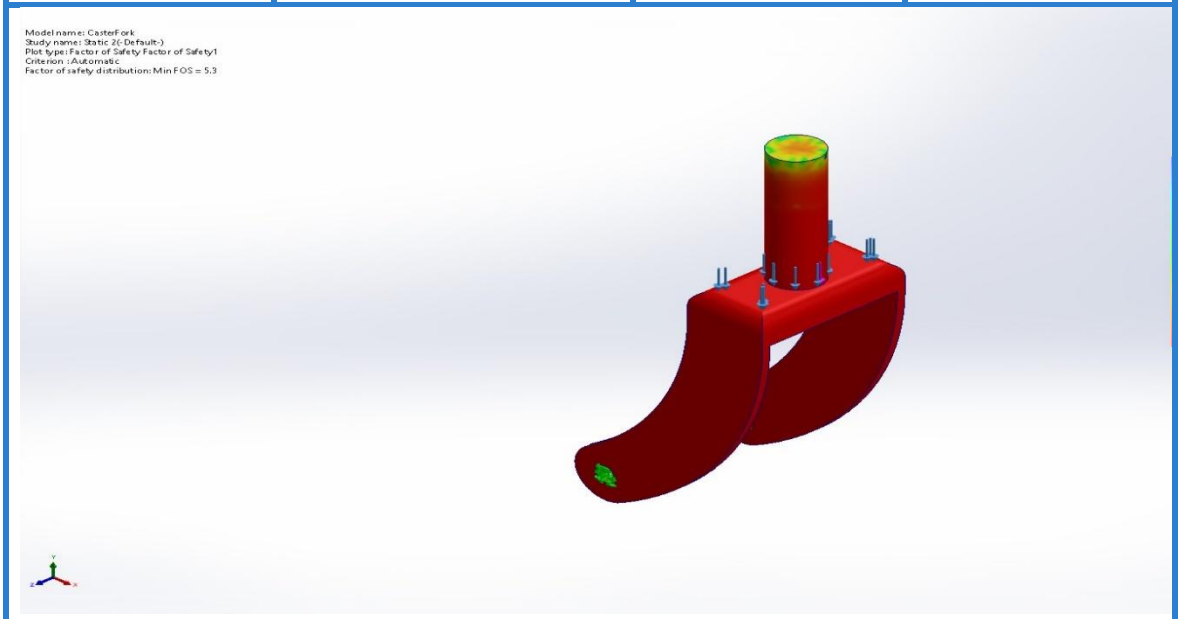
CasterFork-Static 2-Displacement-Displacement1

Name	Type	Min	Max
Strain1	ESTRN: Equivalent Strain	8.593e-12 Element: 5673	4.135e-04 Element: 2962



CasterFork-Static 2-Strain-Strain1

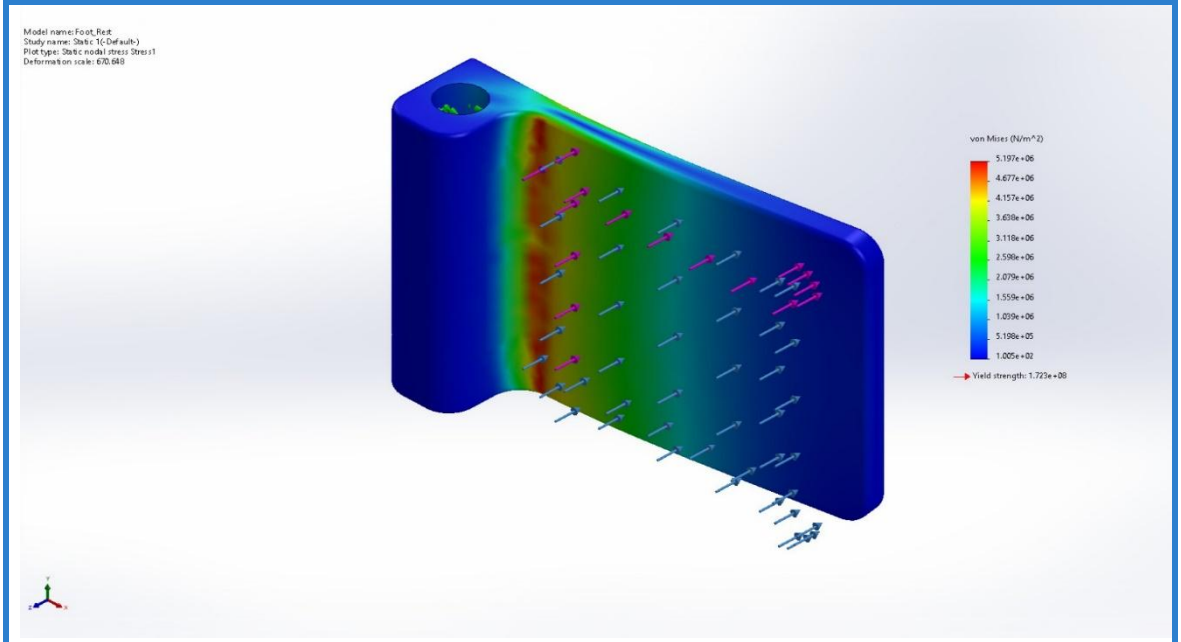
Name	Type	Min	Max
Factor of Safety1	Automatic	5.254e+00 Node: 13000	3.800e+08 Node: 8725



CasterFork-Static 2-Factor of Safety-Factor of Safety1

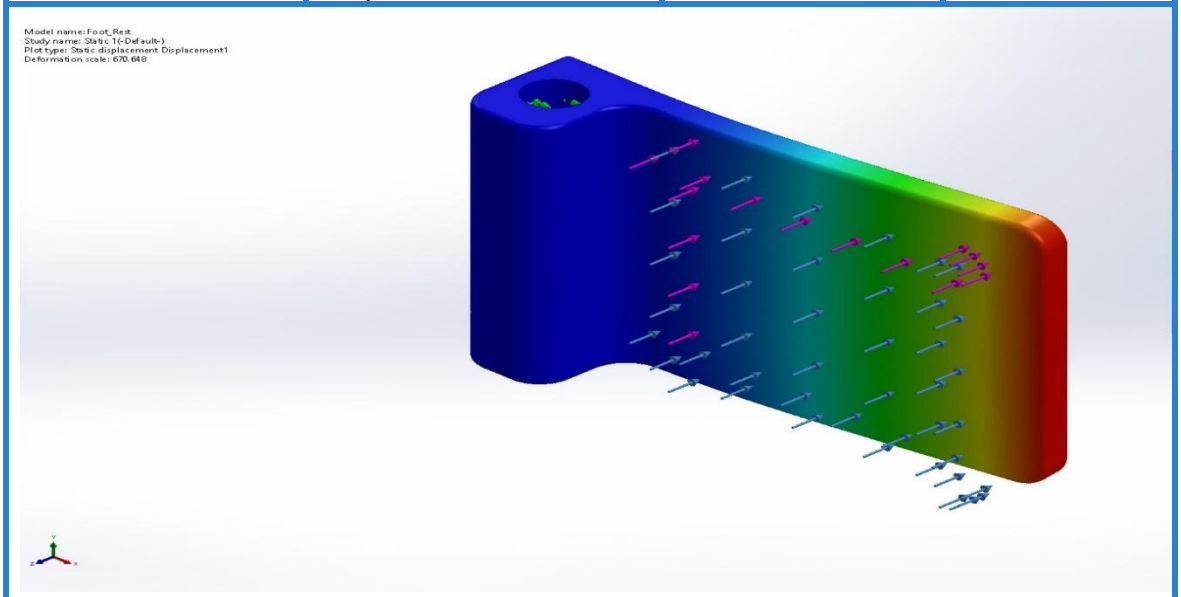
Study Results

Name	Type	Min	Max
Stress1	VON: von Mises Stress	1.005e+02N/m^2 Node: 49926	5.197e+06N/m^2 Node: 46487



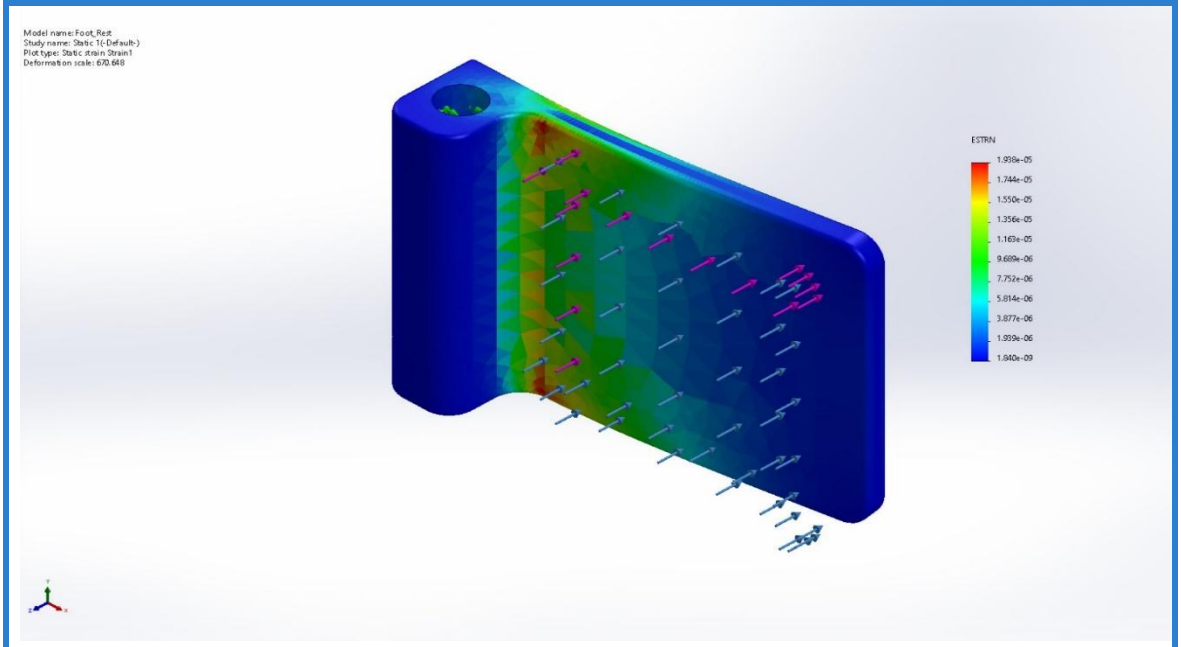
Foot_Rest-Static 1-Stress-Stress1

Name	Type	Min	Max
Displacement1	URES: Resultant Displacement	0.000e+00mm Node: 5	3.032e-02mm Node: 966



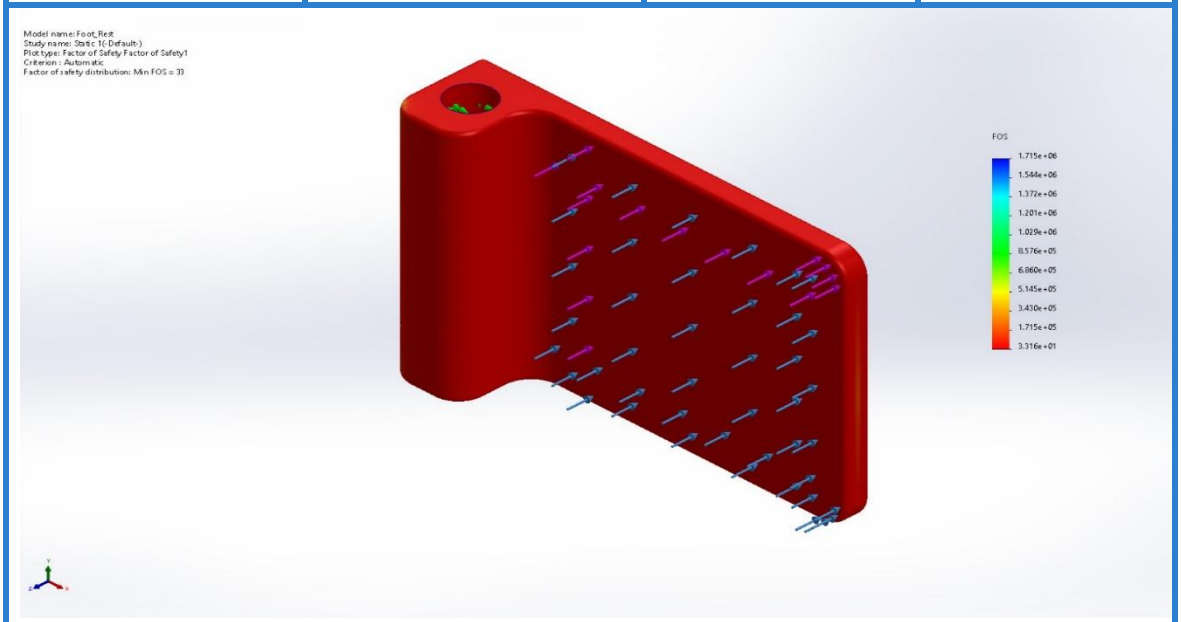
Foot_Rest-Static 1-Displacement-Displacement1

Name	Type	Min	Max
Strain1	ESTRN: Equivalent Strain	1.840e-09 Element: 20644	1.938e-05 Element: 20592



Foot_Rest-Static 1-Strain-Strain1

Name	Type	Min	Max
Factor of Safety1	Automatic	3.316e+01 Node: 46487	1.715e+06 Node: 49926



Foot_Rest-Static 1-Factor of Safety-Factor of Safety1

Chapter 10: Material Selection Process

10.1 Material Selection

Three primary materials were chosen: and Aluminum 6000 series carbon fibre composite, and 304 stainless steel. The Digital Logic Method was used to determine the best material for the wheelchair frame. The material was chosen based on seven predetermined criteria: Cost, weight, toughness, elastic modulus, corrosion resistance, fatigue resistance, and ultimate strength.

Material Properties	**Aluminum 6000 series**	**Stainless Steel(Grade 304)**	**Carbon Fiber Composite**	**Minimum Value Required**	**Maximum Value Required**
Ultimate Strength (MPa)	290	600	650	240	650
Elastic Modulus (GPa)	80	190	170	69	250
Cost (TK)	270	400	2400	270	2400
Toughness (J)	80	200	70	70	200
Weight/M^3	2700	8000	2000	2700	8000

Table 12.1 Material Properties

Some properties were unknown for which a relative evaluation chart was formed in order to give the properties numerical forms.

Properties	**Aluminum 6000 Series**	**Stainless Steel**	**Carbon Fiber Composites**
Corrosion Resistance	4	5	3
Fatigue Resistance	3	4	5

Very High	5
High	4
Medium	3
Low	2
Very Low	1

Table 12.2 Chart for relative evaluation

The **Digital Logic Method** involves determining the relative importance coefficients of selection criteria, scaling the material properties, and ultimately calculating the performance index for each material. The process involves the following steps:

Step 1: The relative importance coefficients of the criteria are determined. In this step, the relative importance of each criterion is compared with every other criterion. For example, in the first column, the relative importance between **Cost** and **Toughness** is evaluated. If Cost is considered more important than Toughness, Cost is assigned a value of 1, while Toughness is assigned a value of 0. This comparison is done for all possible pairs of criteria, and the relative importance coefficient for each criterion is then calculated using a specific formula.

$$\alpha = \frac{Total\ Positive\ decision\ for\ a\ particular\ criteria}{\Sigma(All\ criteria\ Positive\ Decision)}$$

Step 2: The scaled properties for each material were determined and finally their performance indexes

Selection Criteria																						Positive Decisions	Relative Emphasis Co-efficient
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21		
Cost	1	1	1	0	1	1																5	0.238
Toughness	0						0	0	1	1	1											3	0.142
Ultimate Strength		0					1					0	1	1	1							4	0.190
Fatigue Resistance			0					1				1				1	1	1				5	0.238
Weight				1					0				0			0			0	0		1	0.047
Corrosion Resistance					0					0				0			0		1		1	2	0.095
Elastic modulus						0					0				0			0		1	0	1	0.047

Table 12.3 Relative importance of Selection Criterion

were calculated The Scaled properties for the materials were determined using the following formula:

For properties that should have maximum values (strength, toughness…)

$$\beta = \frac{numerical\ value of\ property}{Max\ value\ in\ the\ list} \times 100$$

For properties that should have low values (cost, corrosion …)

$$\beta = \frac{Lowest\ value\ in\ the\ list}{numerical\ value of\ property} \times 100$$

For example, the Ultimate strength of Al was 290 MPa, and the max value in the list was 650 MPa, hence the scaled property for Aluminum on basis of Ultimate Strength was calculated as $\beta = 290/650 \times 100 = 44.62$. Then the weighted score ($\alpha\beta$) was calculated by, $\alpha\beta = 0.19 \times 44.62 = 8.47$. . Similar calculations were repeated for other materials. Finally, the summation of the weighted score ($\alpha\beta$) for each material was determined and the material with highest value was chosen.

		Aluminum 6000 series		Stainless Steel		Carbon Fiber Composites	
Selection Criteria	Weighting Factor, α	Scale Property, β	Weighted Score, αβ	Scale Property β	Weighted Score αβ	Scale Property, β	Weighted Score αβ
Cost	0.238	100	23.8	67.5	16.13	11.25	2.67
Toughness	0.142	40	5.68	100	14.2	35	4.97
Ultimate Strength	0.190	44.62	8.47	92.30	17.53	100	19
Fatigue Resistance	0.238	60	14.28	80	19.04	100	23.8
Weight	0.047	74.07	3.48	25	1.17	100	4.7
Corrosion Resistance	0.095	80	7.6	100	9.5	60	5.7
Elastic modulus	0.047	42.10	1.97	100	4.7	89.47	4.20
			$\Sigma\alpha\beta$ =65.29		$\Sigma\alpha\beta$ =82.27		$\Sigma\alpha\beta$ =65.04

Table 12.4 Determination of Performance Index of materials

The Stainless Steel was found to have the Performance Index (γ), 82.27, hence, it was selected as the preferred material for wheelchair frame.

Chapter 11: Manufacturing Process Selection

One of the primary challenges encountered by design and manufacturing professionals is determining the most suitable production approach for creating a part. This decision is complicated by the wide variety of raw materials and manufacturing processes available, each with distinct implications for both design and production. A summary of the key selection criteria is outlined below. The application requirements form the foundation of the process selection. When selecting an appropriate process, it is essential to consider factors such as the part's dimensions, mechanical properties, cost efficiency, and production rate.

The primary objective is to utilize machinery, labor, equipment, and chemical processes to convert raw materials or components into finished products. Choosing the correct process for a product part is essential, as it directly influences the overall design of the product

Certain procedures are chosen for the product that are:

1. Gas Welding
2. Arc Welding

Selection Criteria	**Decision Numbers**															**Positive Decisions**	**Relative Emphasis Coefficient,α**
	1	2		4	5	6	7	8	9	I 0	11	12	13	14	15		
Low Cost	0	1	0	1												2	0.13
Dimensional Accuracy	1					1	1	1	0							4	0.27
Processing Time		0				0					1	0				1	0.07
Skill Required for Operation			1				0			1			1	1		4	0.27
Waste Generation				0				0			0		0		1	1	0.07
Surface Finish					1				1			1		0	0	3	0.2
Total																**15**	**1**

3. MIG welding

Table 13.1 Priority decisions of different criteria

Selection Criteria	Gas Welding	Arc Welding	MIG Welding
Low Cost	5	4	2
Dimensional Accuracy	4	3	4
Processing Time	3	2	4
Skill Required For Operation	4	3	2
Waste Generation	5	2	3
Surface Finish	4	3	4

Table 13.2 Chart for Evaluation (Process)

Very High	5
High	4
Medium	3
Low	2
Very Low	1

Table 13.3 Chart for Relative Evaluation Values (Process)

After that, we measured the scaling factors.

For properties that should have maximum values

β = (Numerical value of property/ max value in the list) *100

For properties that should have minimum values

Now we calculate the performance index of each material

$\gamma = \Sigma\alpha\beta$

The process has the highest γ, performance index will be the best for our product.

Criteria	α	Type	Gas Welding	β (Gas)	α×β (Gas)	Arc Welding	β (Arc)	α×β (Arc)	MIG Welding	β (MIG)	α×β (MIG)
Low Cost	0.13	Min	5	100	13	4	80	10.4	2	40	5.2
Dimensional Accuracy	0.27	Max	4	100	27	3	75	20.25	4	100	27
Processing Time	0.07	Min	3	75	5.25	2	50	3.5	4	100	7
Skill Required	0.27	Min	4	100	27	3	75	20.25	2	50	13.5
Waste Generation	0.07	Min	5	100	7	2	40	2.8	3	60	4.2
Surface Finish	0.2	Max	4	100	20	3	75	15	4	100	20
Total (γ)					99.25			72.2			76.9

Table 13.4 : Calculation of the Performance Index (Process)

Result: We will use **gas welding** for our manufacturing process

Chapter 12: Cost Calculations

12.1 Product structure/Bill of Materials(BOM)

The parts or components requirements of the final product which is to be manufactured is usually presented in the form of a product structure or bill of material . It is a listing of all components (sub assemblies and materials) that go into an assembled item. It frequently includes the part numbers and quantity required for assembly.

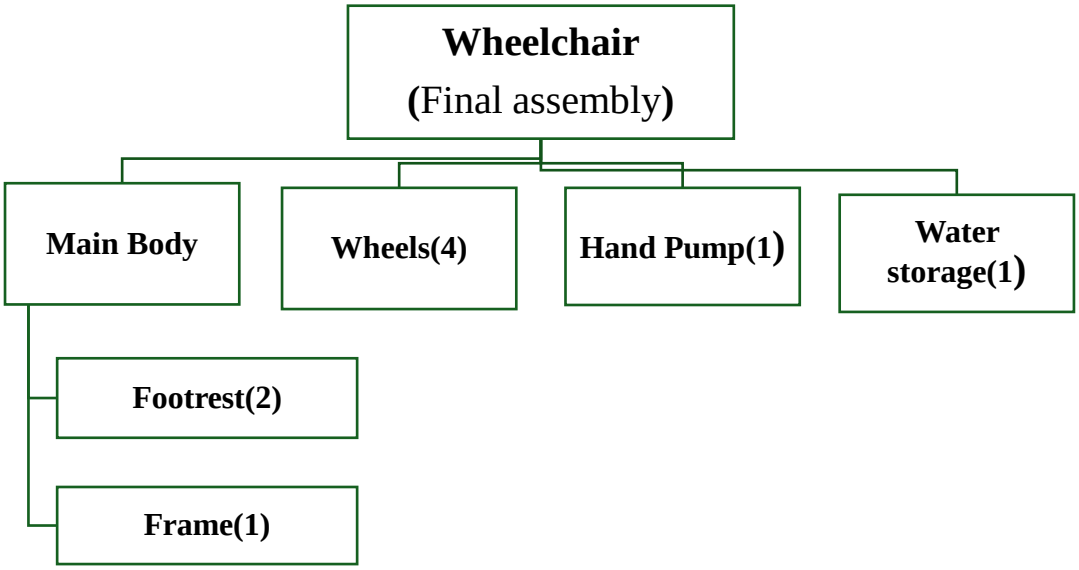


Figure 14.1 Bill of materials of wheelchair

Details of bill of materials for our wheelchair is given below-

SL No.	Component	Material	Unit Cost	No of Unit	Cost	% of Total Cost
1	Metal Frame	SS	3125	1	3125	43.19%
2	Commode	plastic	475	1	475	6.57%
3	Footrest	steel	255	2	510	7.05%
4	Primary Wheels	steel	895	2	1790	24.74%
5	Front Wheel	steel	322	2	644	8.90%
6	Hand Pump	plastic	220	1	220	3.04%
7	Water storage	plastic	471	1	471	6.51%
					7235	

Table 14.1 BOM Table

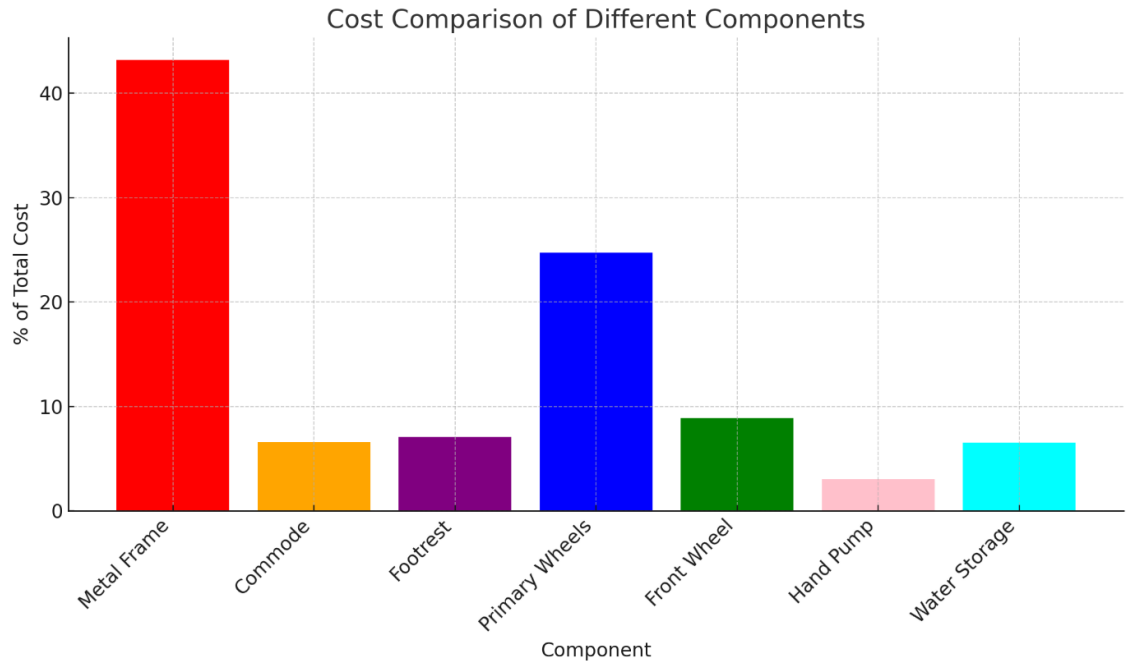


Figure 14.2 Cost Comparison of different element

12.2 Lifecycle Cost

Lifecycle Cost refers to the total cost of owning, operating, and maintaining a product or asset over its entire lifespan. It includes initial purchase costs, operating costs, maintenance costs, and disposal or end-of-life costs. This concept helps organizations make more informed decisions by evaluating the long-term financial impact rather than just the initial cost.

	Fixed Cost (TK.)						**Variable Cost (TK.)**
	1st Year	**2nd Year**	**3rd Year**	**4th Year**	**5th Year**	**Lifecycle**	**Per Unit Cost**
R & D							
Design	35,000	-	-	-	-	35,000	
Improvements	15,000	10,000	8,000	5,000	7,500	45,500	
Total R & D	50,000	10,000	8,000	5,000	7,500	80,500	
Facilities Expenses							
Land Cost	842,772	842,772	842,772	842,772	842,772	4,213,860	
Depreciation	5,000	5,000	5,000	5,000	5,000	25,000	
Total Facilities Exp.	847,772	847,772	847,772	847,772	847,772	4,238,860	
Production							
BOM							7,235

Labor Cost							420
Maintenance	61,223	69,777	82,102	88,897	92,129	394,128	
Utilities	200,122	250,213	350,706	400,233	350,127	1,551,401	
Insurance	60,000	60,000	60,000	60,000	60,000	300,000	
Total Production Cost	321,345	379,990	492,808	549,130	502,256	2,245,529	
Marketing							
Awareness	120,000	90,000	70,000	50,000	40,000	370,000	
Accessibility	50,000	40,000	30,000	20,000	10,000	150,000	
Total Marketing Cost	170,000	130,000	100,000	70,000	50,000	520,000	
Distribution							
Logistic Management	59,223	55,107	48,321	38,139	40,278	241,068	
Shipping Cost							175
Total Distribution Cost	59,223	55,107	48,321	38,139	40,278	241,068	
Customer Service							
Repairing Workshop	81,239	85,326	78,476	82,377	80,709	408,127	
Free 1st Month Adjustment	15,123	12,678	11,788	9,344	10,778	59,711	
Total Customer Service Cost	96,362	98,004	90,264	91,721	91,487	467,838	
Additional Overheads							
Administration	320,344	398,378	620,434	788,234	802,355	2,929,745	
Patents	65,000					65,000	
Lawsuits	45,301	20,222	33,325	38,766	29,977	167,591	
Software License	3,000	3,000	6,000	6,000	6,000	24,000	
Total Overhead Cost	433,645	421,600	659,759	833,000	838,332	3,186,336	
End of Life Cycle Project							
Decommissioning					54,871	54,871	
Waste management					46,347	46,347	
Total EOC cost					101,218	101,218	
Total Accumulated Cost	1,978,347	1,942,473	2,246,924	2,434,762	2,478,843	11,081,349	7,830

Table 14.2 Life Cycle Cost

12.3 Sales Scenario

Year	Sales(unit)
1	173
2	409
3	1526
4	1154
5	976
Total Unit Produced	4238

Table 14.3 Sales Scenario

12.4 Cost Calculation

• Variable Cost:

1. Bill of Materials: Total per unit cost =7,235 Taka

2. Lifecycle Cost: Total accumulated variable cost = 7,235*4,238 = 30,661,930 Taka

• Fixed Cost:

1. Lifecycle Cost: Total accumulated fixed cost = 11,081,349 Taka

• Total Cost:

Fixed Cost + Variable Cost = 41,743,279 Taka.

• Per Unit Cost:

Total Cost ÷ Total Unit Production = 41,743,279 /4,238 = 9849.7591 Taka

12.5 Break Even Analysis

Let,

BEPx = Break Even Point in Units.

BEP$ = Break Even Point in Takas.

S = Selling price per unit = 12,500 Taka

Q = Total number of units produced = 4,238 units

TR = Total Revenue = $(s \times Q)$ = Total Sales = 52,975,000 Taka

F = Fixed Cost = 11,081,349 Taka

V = Variable Cost = 7,235 Taka/unit

TC = Total Cost = F + VQ = 41,743,279 Taka.

Break Even Point occurs when TR - TC.

BEPx =F/(S-V) =11,081,349 /(12,500-7,235) = 2,104.72 Units.

BEP$ = BEPx × S = 2,104.72 *12,500 = 26,308,995 Taka.

Profit = TR – TC

= 52,975,000 - 41,743,279 =11,231,721

The Break-Even Analysis graph is provided below:

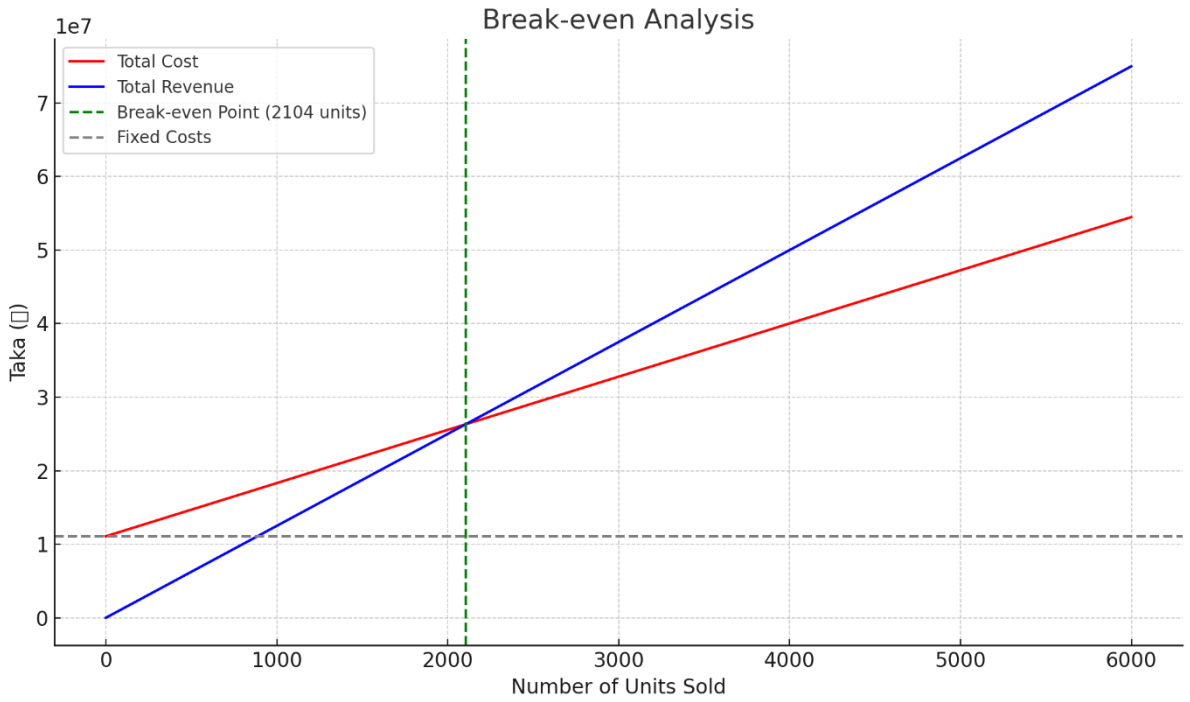


Figure 14.3 Break Even Analysis

Chapter 13: Conclusion

The design and development of the "Wheelchair-Bed that Converts to a Bed with Self-Cleaning Feature for Improved Mobility and Comfort for Immobile Patients" presents a significant step forward in the realm of assistive devices for individuals with limited mobility. Through extensive market research and customer surveys, it was evident that existing wheelchair models do not adequately address key user concerns regarding comfort, accessibility, and hygiene. The proposed wheelchair-bed aims to fill these gaps by incorporating essential features such as a defecation system, a reclining mechanism for enhanced comfort, and a cleaning system, all of which were identified as crucial by the target audience.

By applying the Kano Model, we successfully identified and prioritized the most important customer requirements, including features like the 180-degree reclining capability, defecation system, and self-cleaning functionality, ensuring that our design resonates with user needs. The competitor analysis also highlighted the lack of similar innovative features in the market, providing our product with a distinct competitive advantage.

Incorporating the findings from the Quality Function Deployment (QFD) process, we ensured that the design of this wheelchair-bed not only meets but exceeds customer expectations, offering practical, user-friendly solutions for immobile patients. Additionally, our stress analysis, material selection, and manufacturing process choices were carefully made to ensure durability, cost-effectiveness, and reliability of the final product.

Overall, the development of this convertible wheelchair-bed is positioned to make a significant impact on the lives of immobile patients, enhancing their quality of life by improving both their mobility and comfort. This product addresses an underserved market segment and offers a compelling solution to the challenges faced by individuals with limited mobility, caregivers, and healthcare facilities alike.

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